Lecture 18:

Intro to Animation

Computer Graphics and Imaging
UC Berkeley CS184

Principles of Animation

CS184

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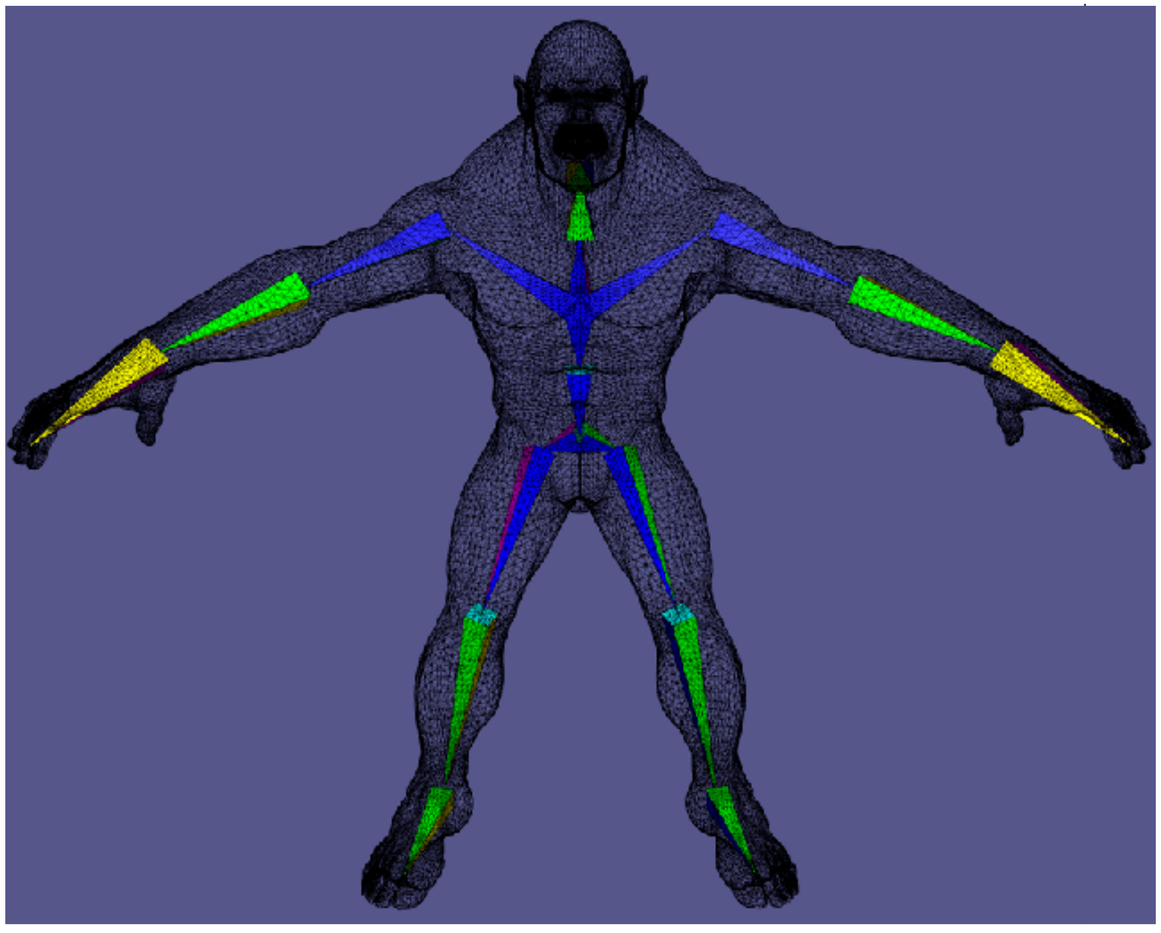
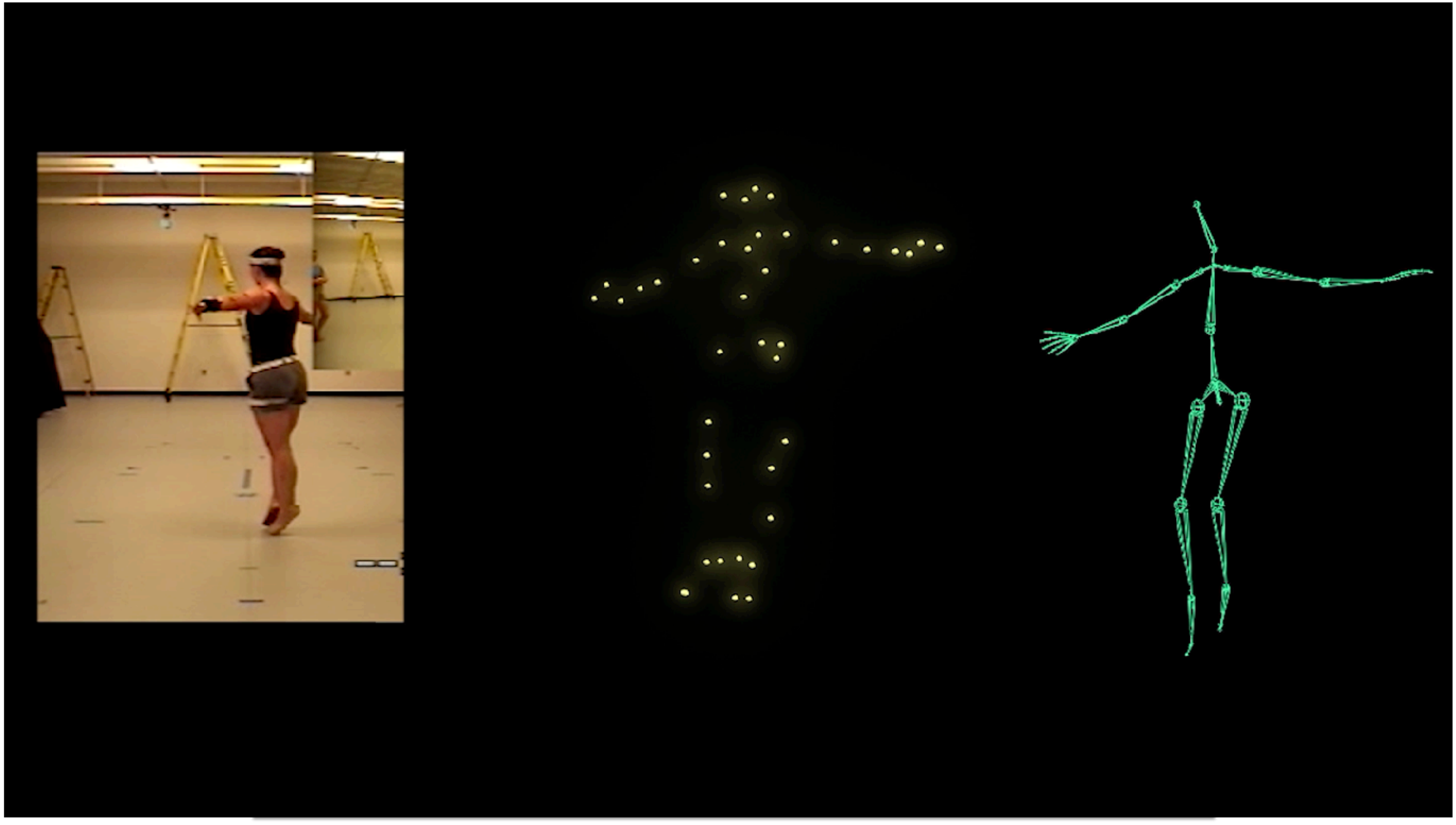
Presented by David McAllister

Rigging & Skinning



Courtesy of libigl / Alec Jacobson

Motion Capture



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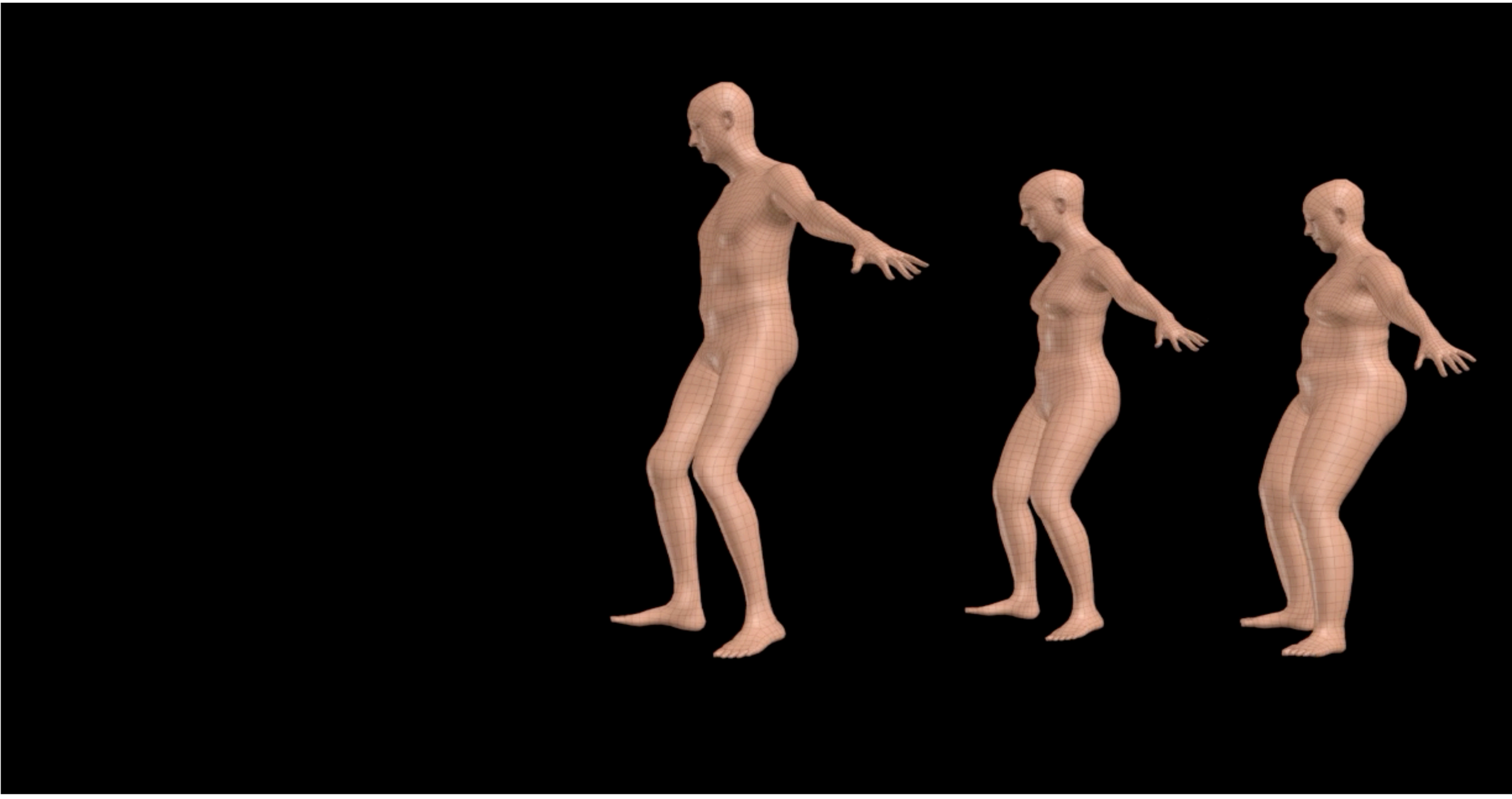
Physical Simulation: Cloth

Parametric Models



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Loper et al. SIGGRAPH Asia 2014

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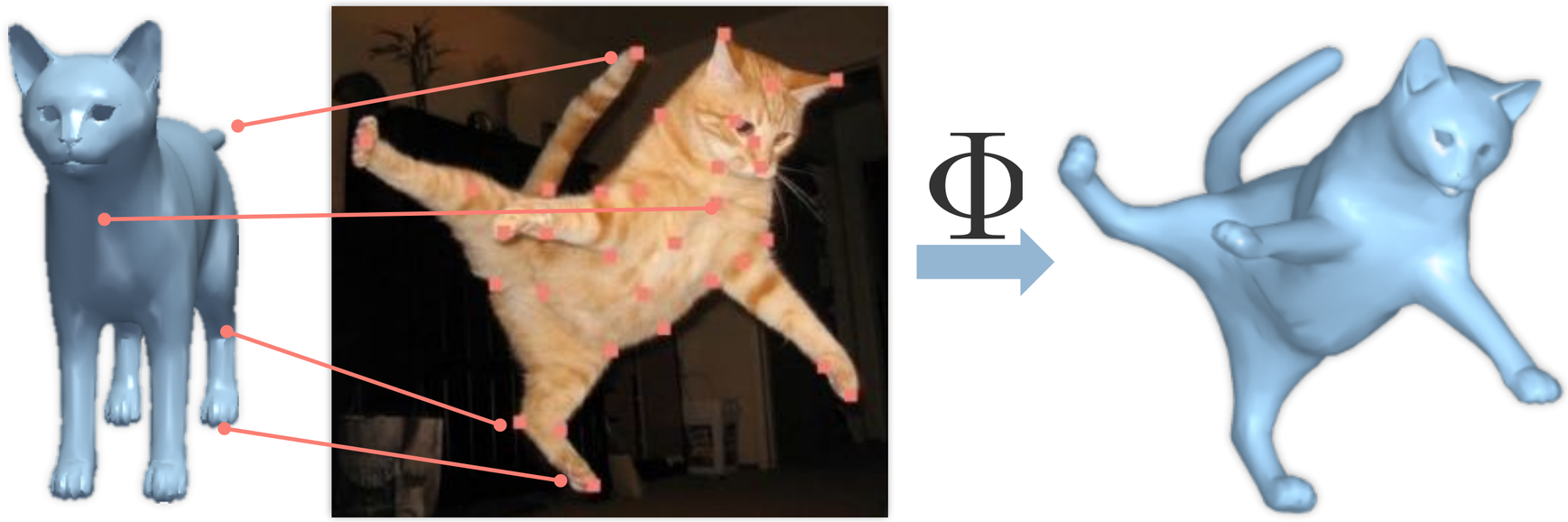
Later: Deformation from Images



Kanazawa et al. Eurographics '16

Zuffi et al. CVPR 2017

Animation

"Bring things to life"

- Communication tool
- Aesthetic issues often dominate technical issues

An extension of modeling

- Represent scene models as a function of space

Output: sequence of images that when viewed sequentially provide a sense of motion

- Film: 24 frames per second
- Video: 30 fps
- Virtual reality: 90 fps

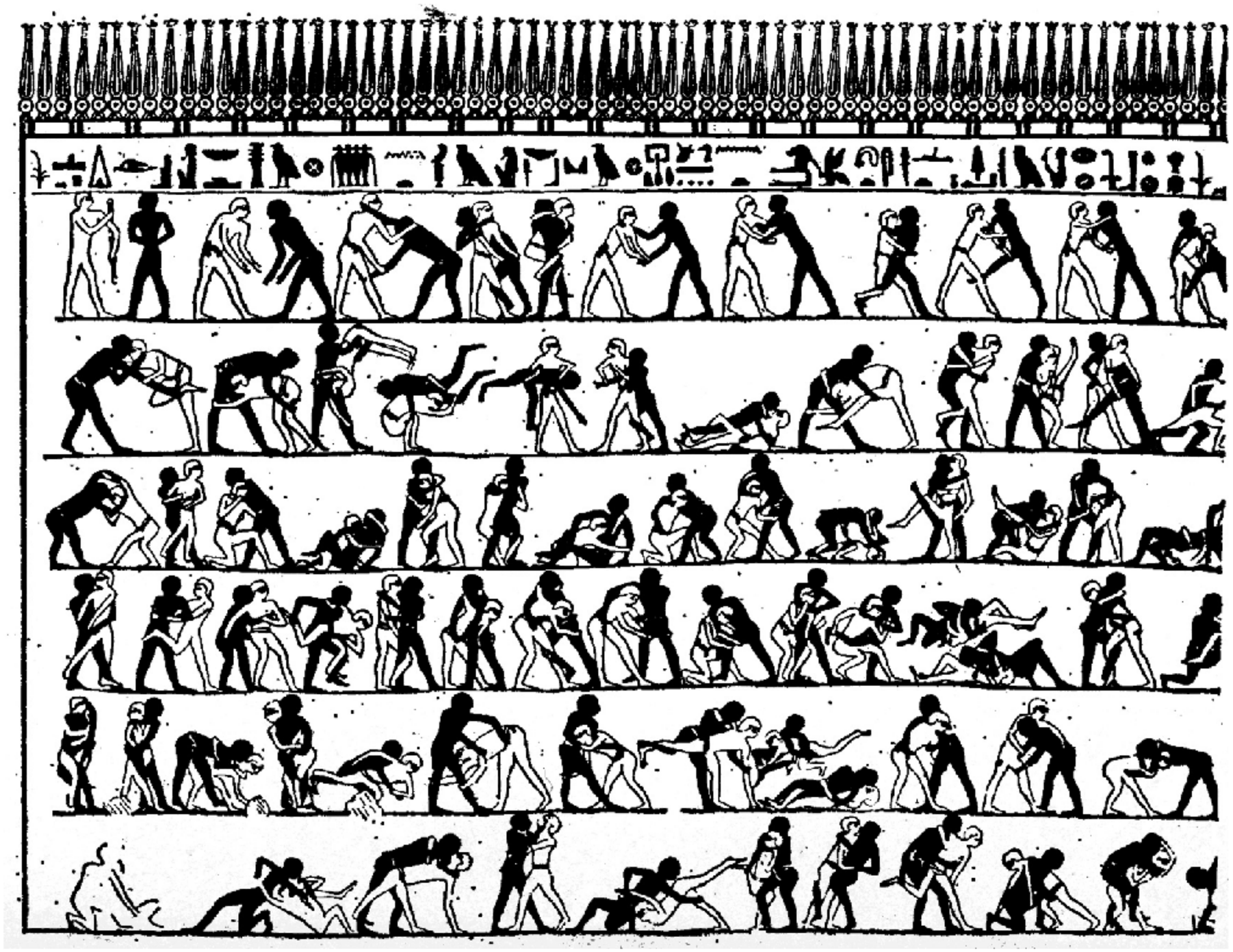
Historical Points in Animation

(slides courtesy Keenan Crane)

First Animation

(Shahr-e Sukhteh, Iran 3200 BCE)

History of Animation



(tomb of Khnumhotep, Egypt 2400 BCE)

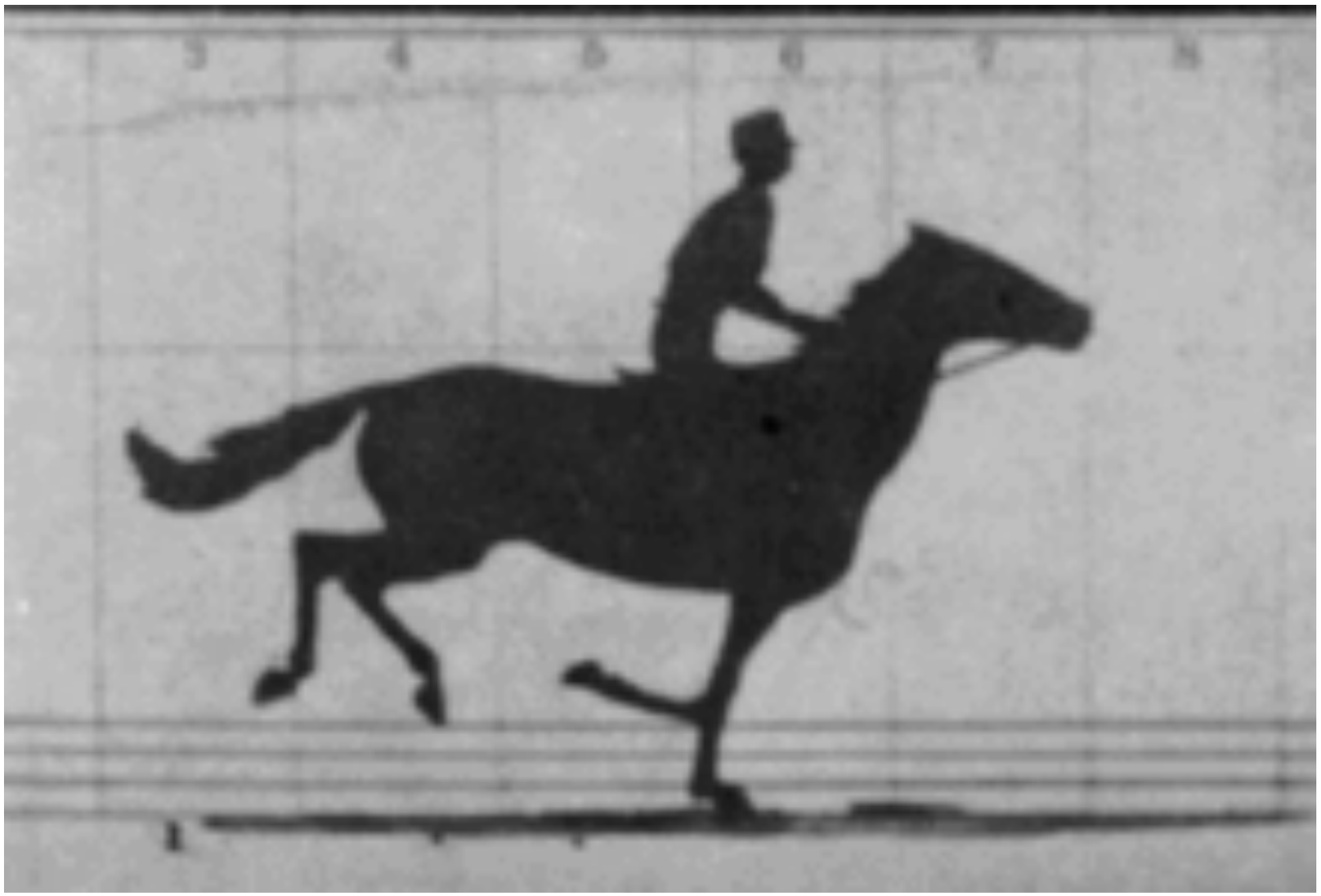
History of Animation

(Phenakistoscope, 1831)

First Film

Originally used as scientific tool rather than for entertainment

Critical technology that accelerated development of animation

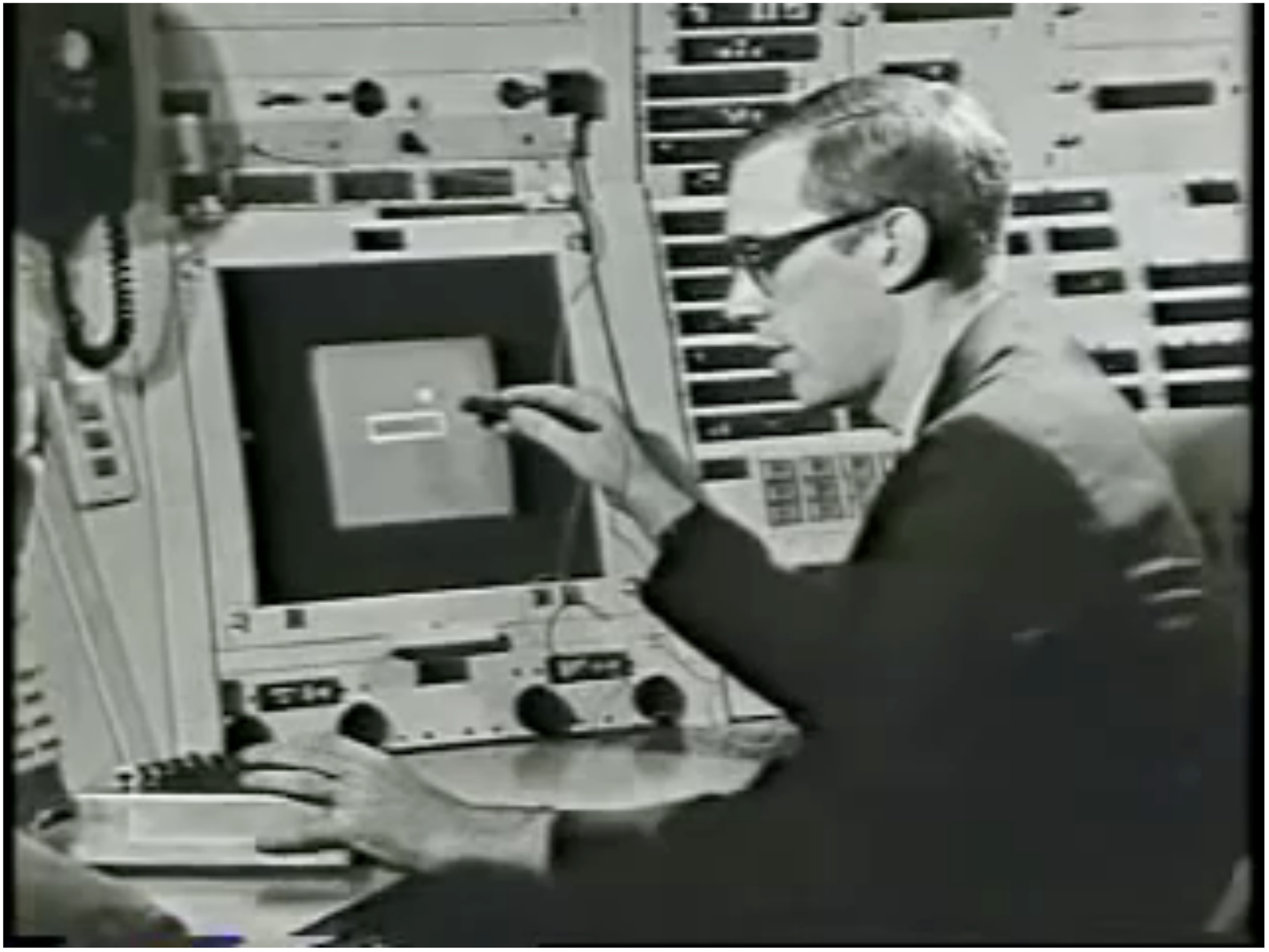


Edward Muybridge, *"Sallie Gardner"* (1878)

First Hand-Drawn Feature-Length Animation

Disney, “Snow White and the Seven Dwarfs” (1937)

First Digital-Computer-Generated Animation



Ivan Sutherland, “Sketchpad” (1963) – Light pen, vector display

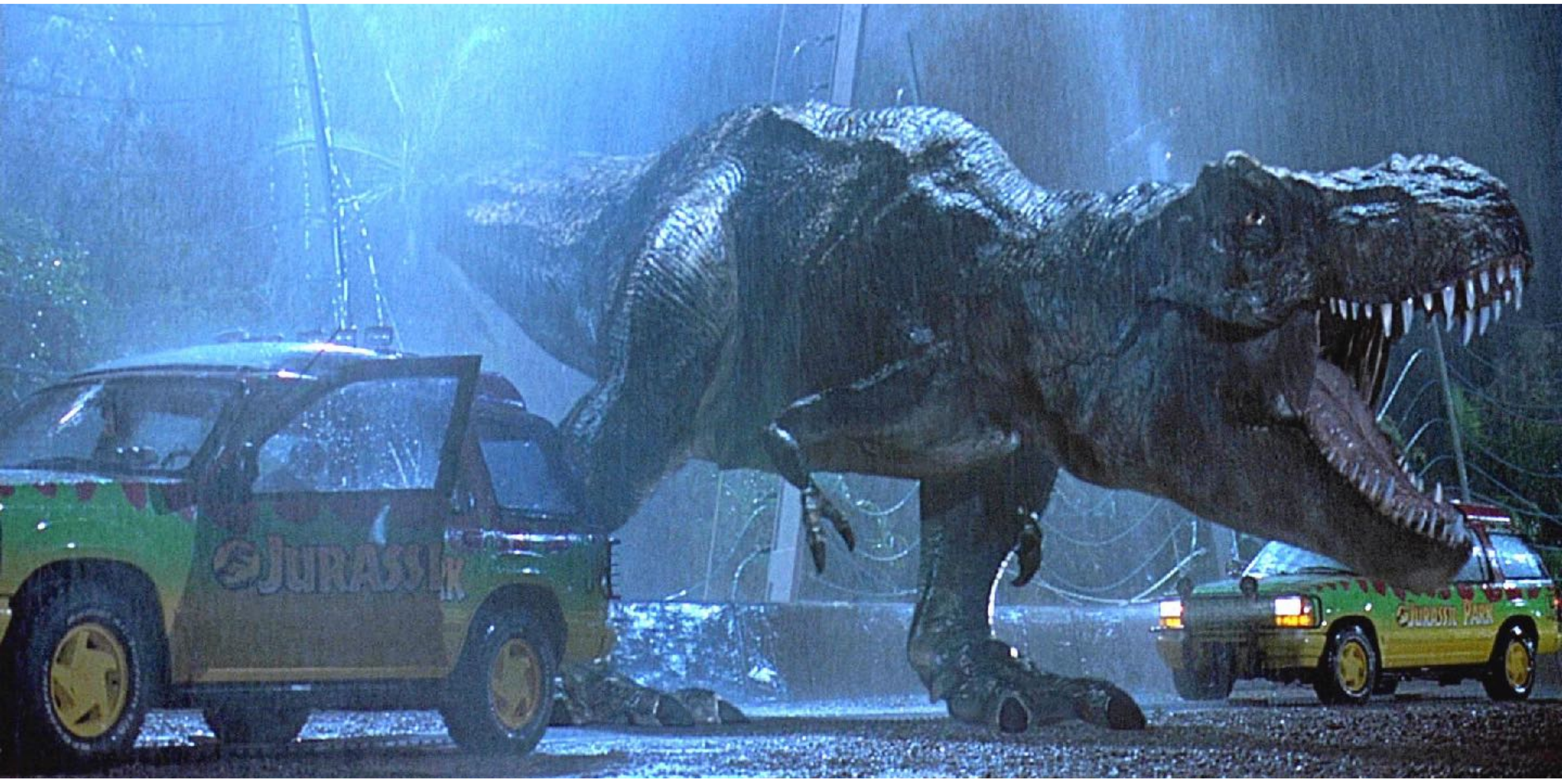
Early Computer Animation

Nikolay Konstantinov, “Kitty” (1968)

Early Computer Animation

Ed Catmull & Frederick Parke, “Computer Animated Faces” (1972)

Digital Dinosaurs!

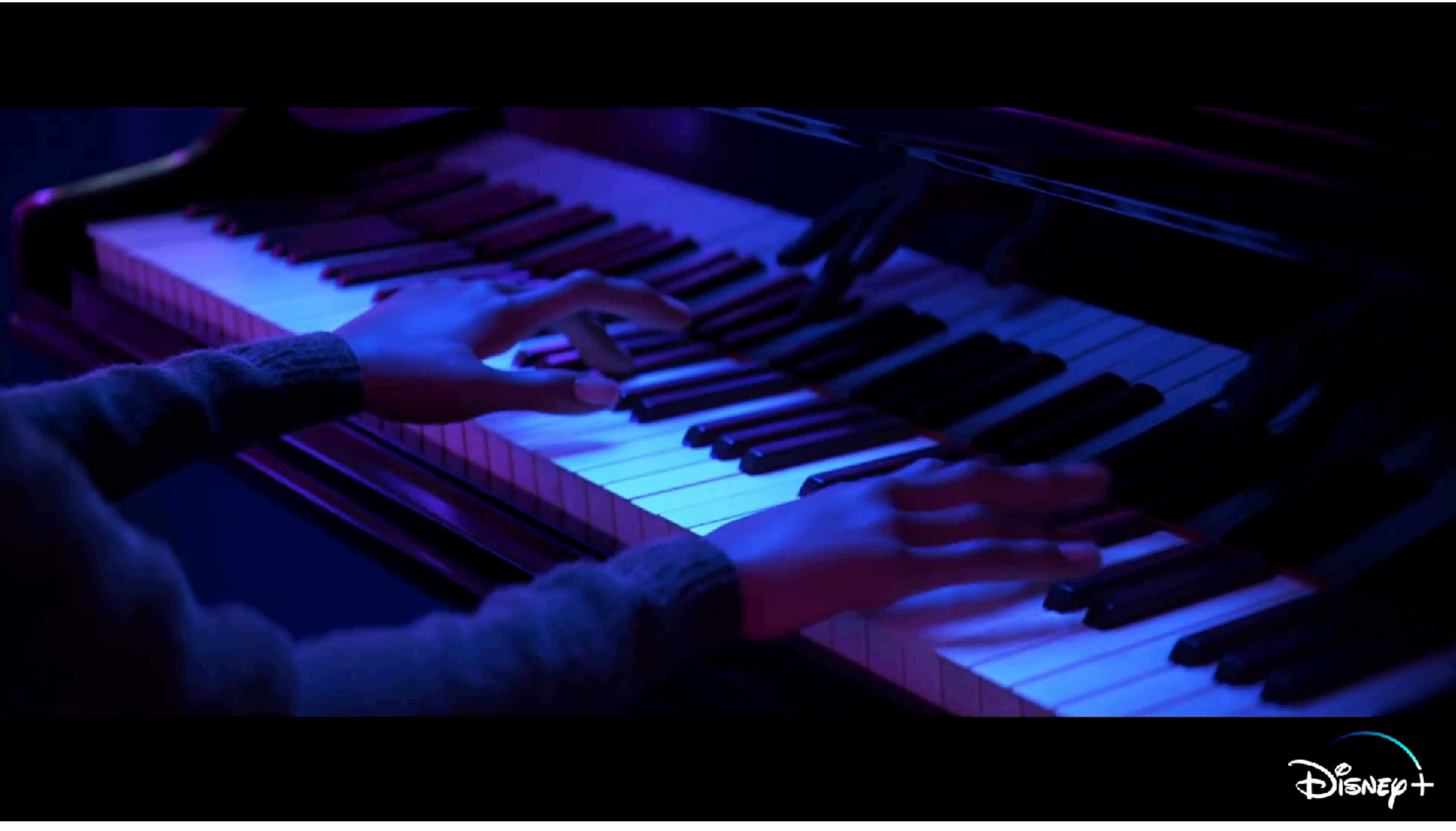


Jurassic Park (1993)

First CG Feature Film

Pixar, “Toy Story” (1995)

Computer Animation - Present Day



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Disney/Pixar Soul (2020)

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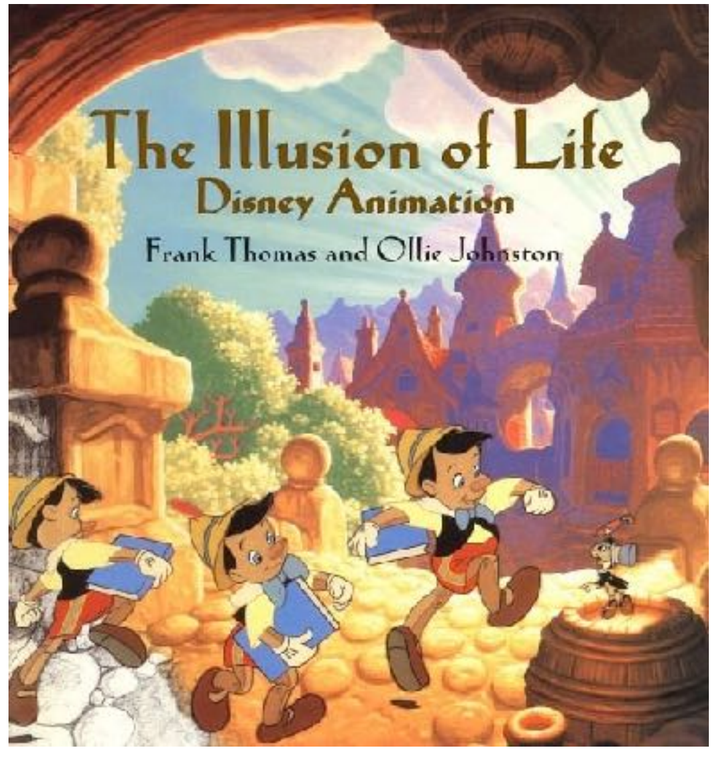
Animation Principles

(slides courtesy Mark Pauly)

Animation Principles

From

- "Principles of Traditional Animation Applied to 3D Computer Animation" - John Lasseter, ACM Computer Graphics, 21(4), 1987



In turn from

- "The Illusion of Life"
  Frank Thomas and Ollie Johnson

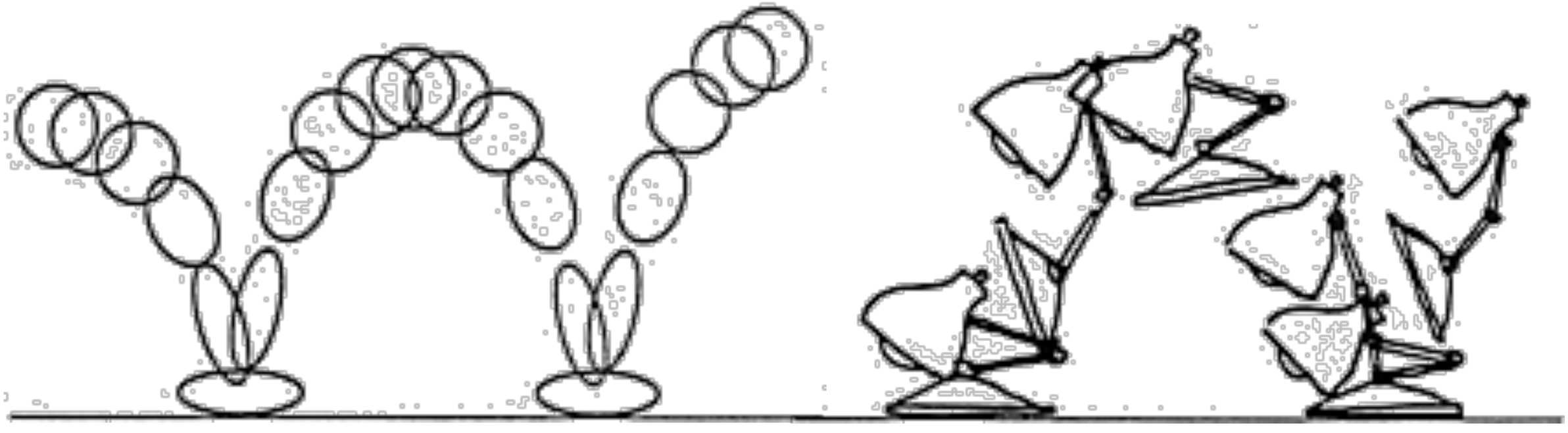
Same for 2D and 3D

http://www.siggraph.org/education/materials/HyperGraph/animation/character_animation/principles/prin_trad_anim.htm

Squash and Stretch

Refers to defining the rigidity and mass of an object by distorting its shape during an action.

Shape of object changes during movement, but not its volume.

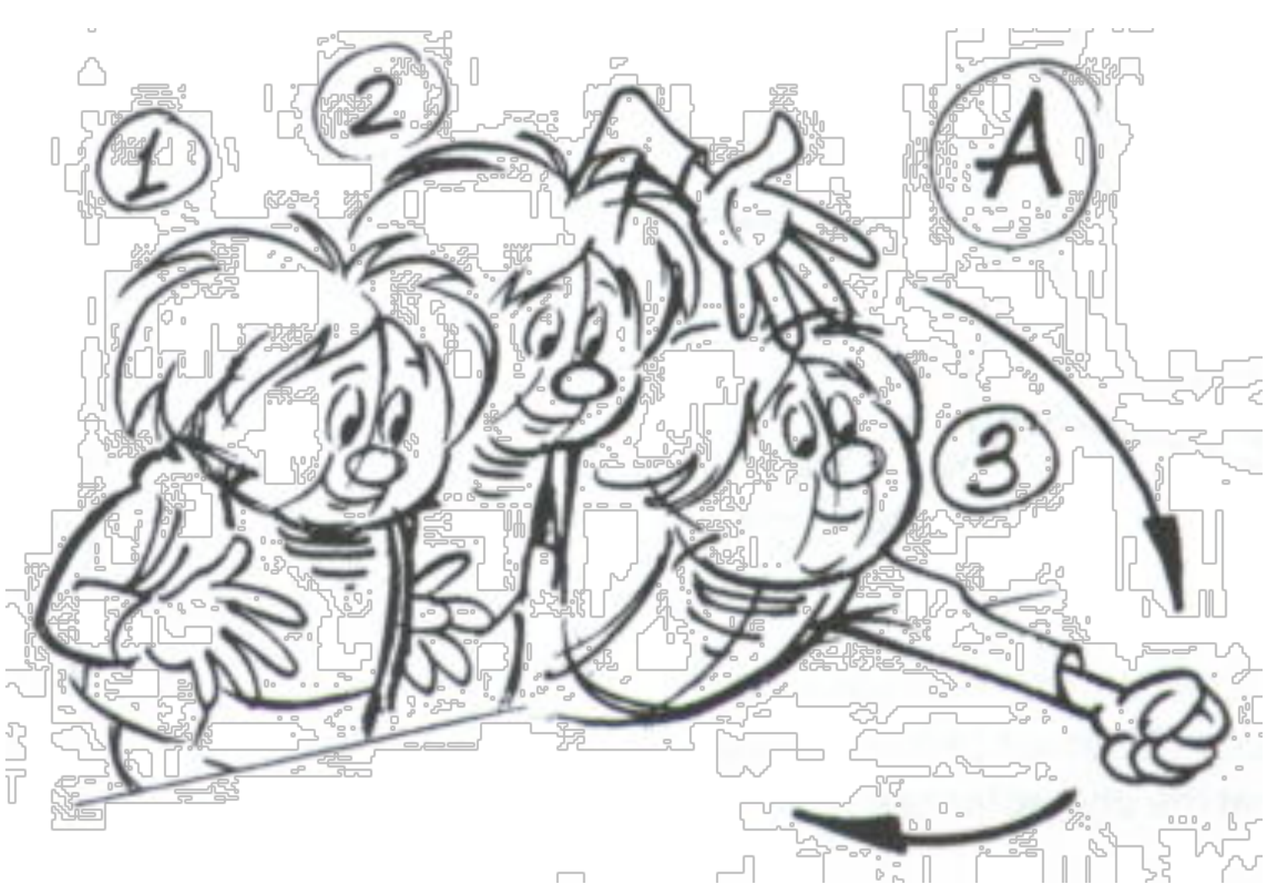


Anticipation

Prepare for each movement

For physical realism

To direct audience's attention



Timing for Animation, Whitaker & Halas

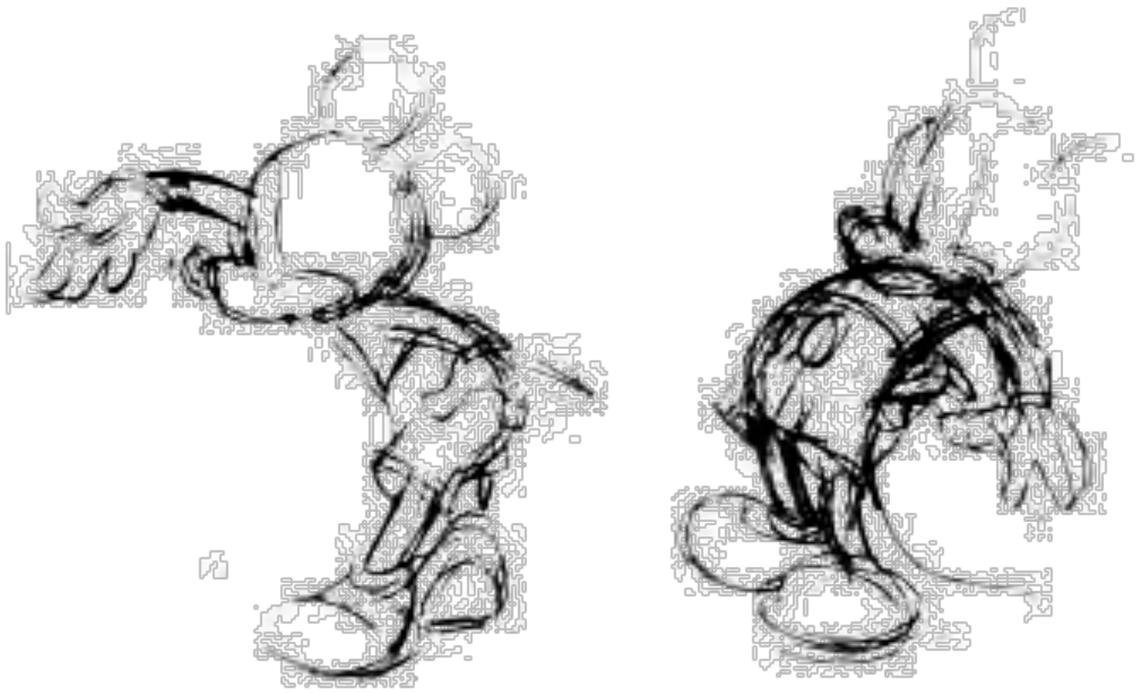
Staging

Picture is 2D

Make situation clear

Audience looking in right place

Action clear in silhouette



Disney Animation: The Illusion of Life

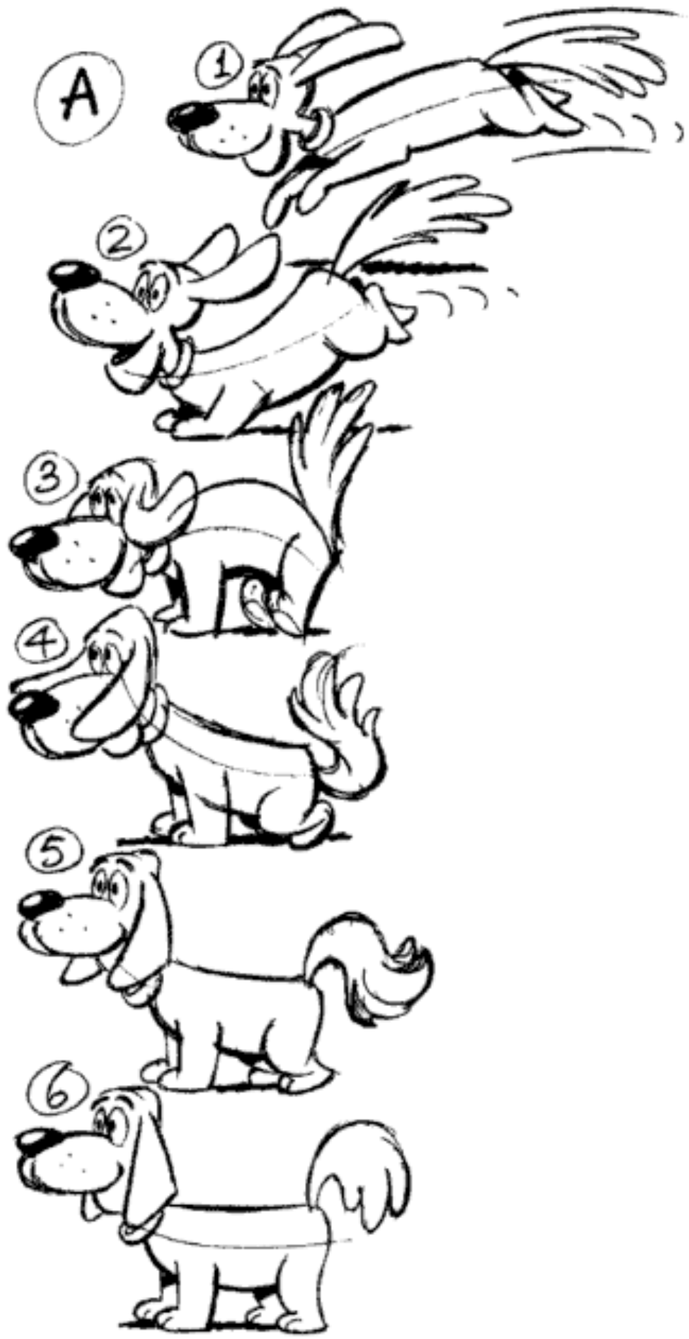
Follow Through

Overlapping motion

Motion doesn’t stop suddenly

Pieces continue at different rates

One motion starts while previous is finishing, keeps animation smooth

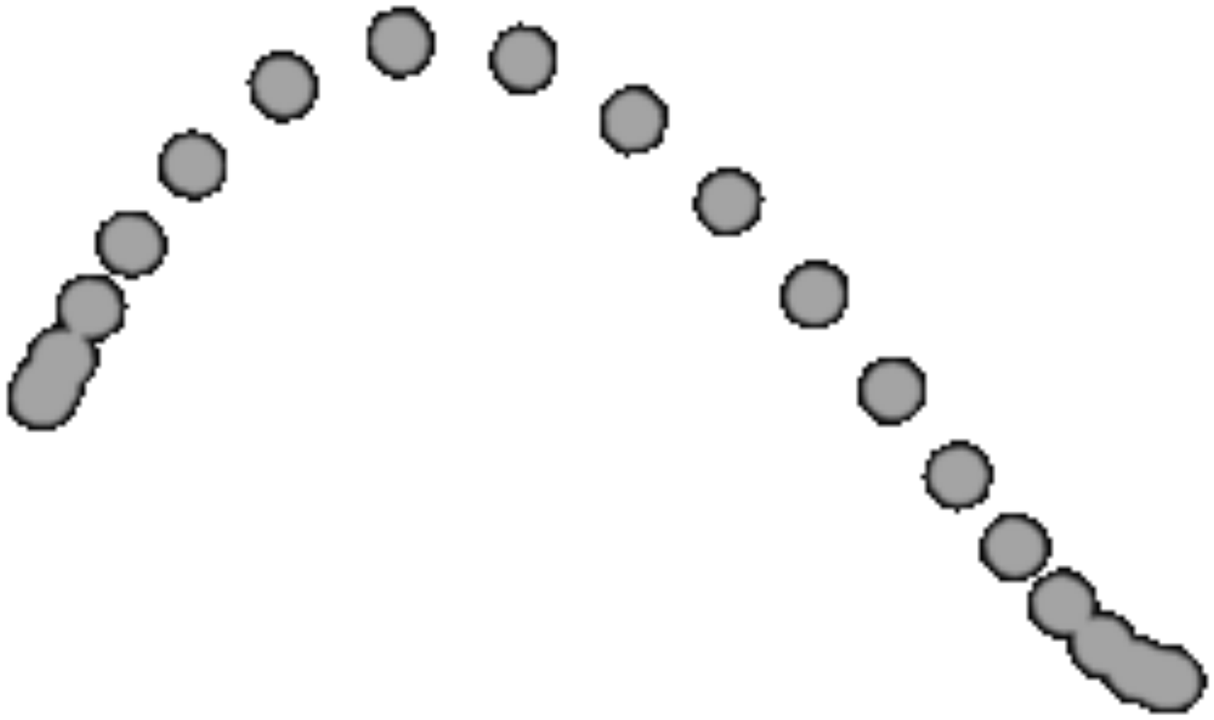


Timing for Animation, Whitaker & Halas

Ease-In and Ease-Out

Movement doesn’t start & stop abruptly.

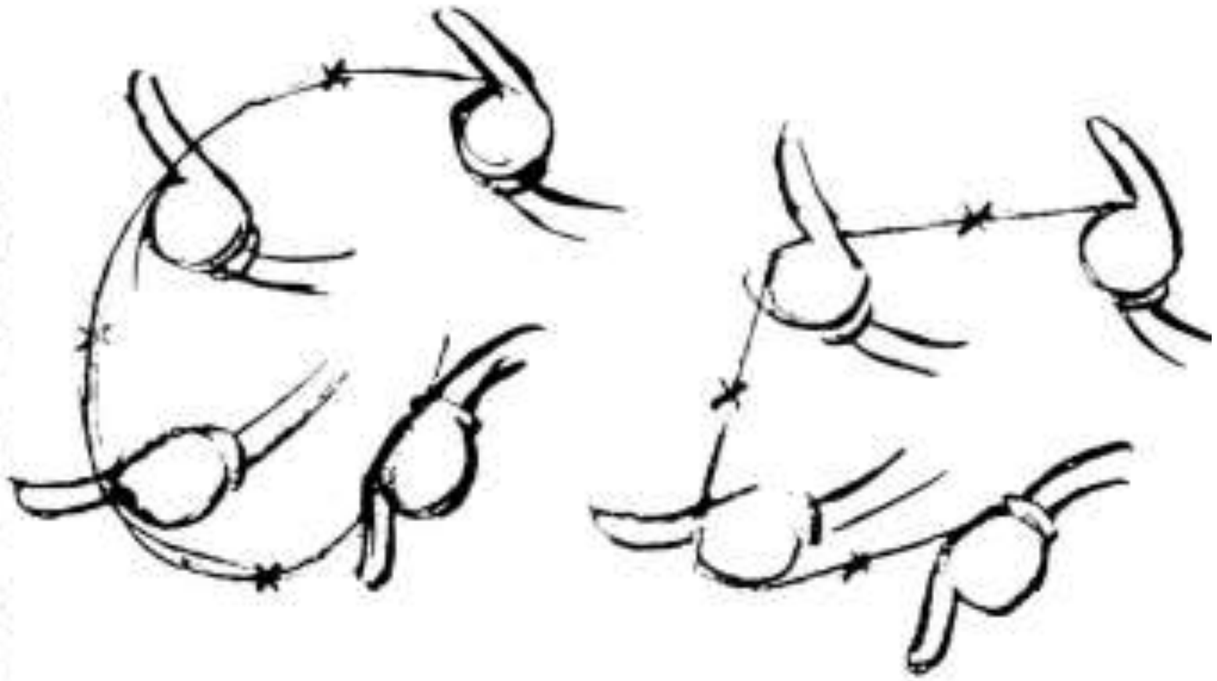
Also contributes to weight and emotion



Arcs

Move in curves, not in straight lines

This is how living creatures move



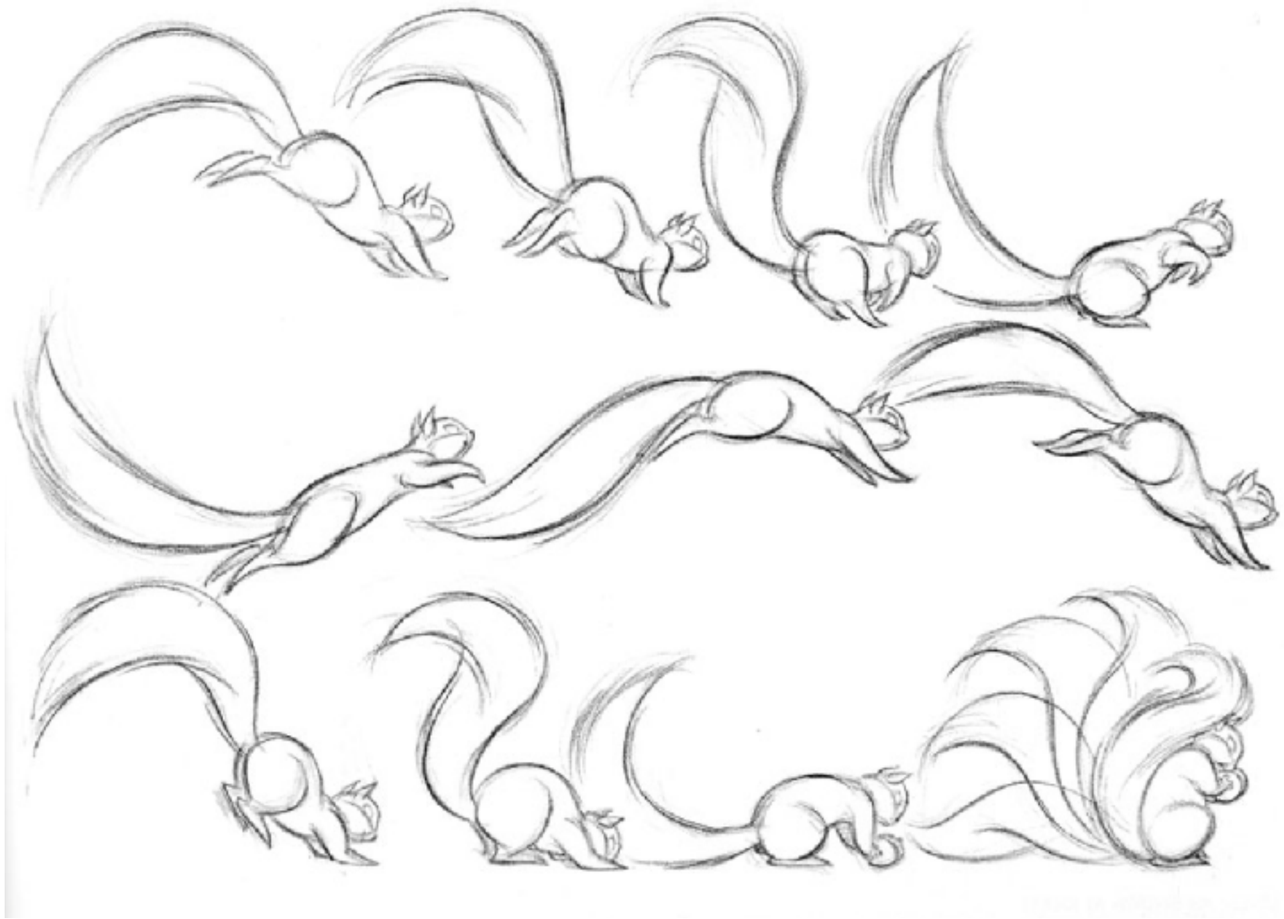
Disney Animation: The Illusion of Life

Secondary Action

Motion that results from some other action

Needed for interest and realism

Shouldn’t distract from primary motion

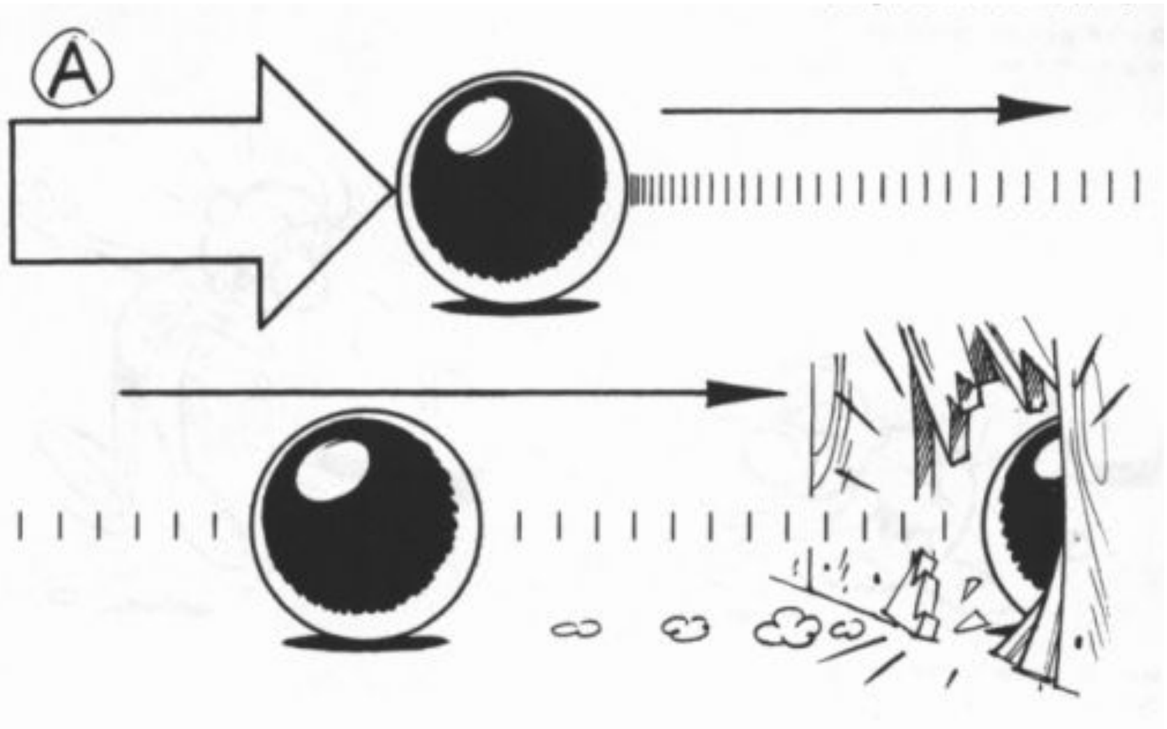


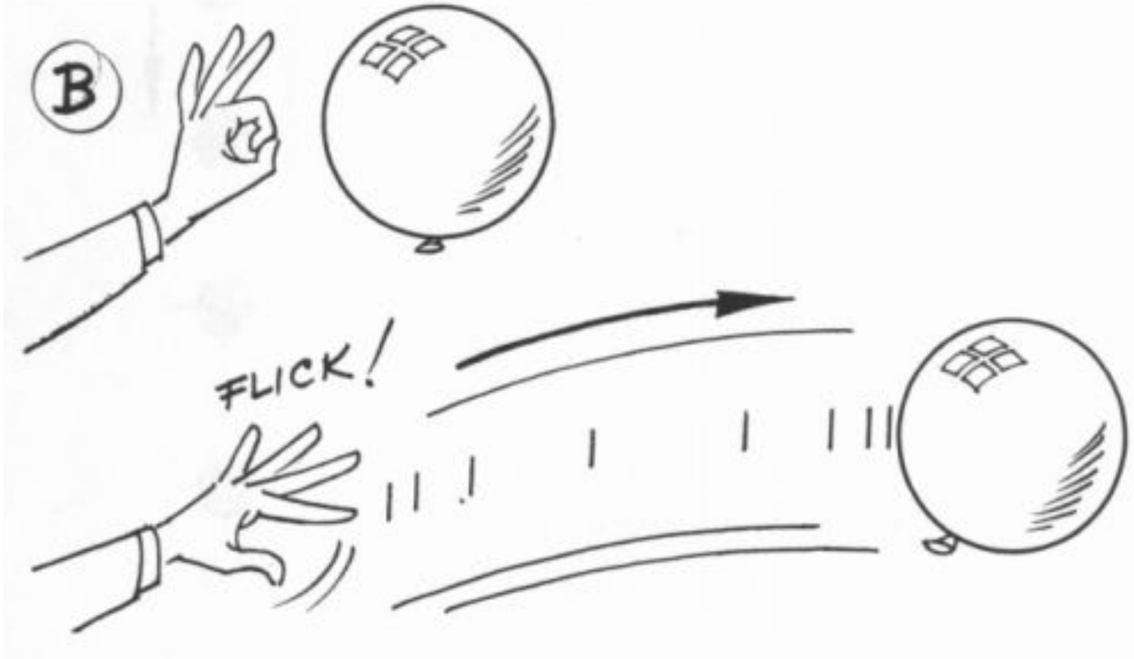
Cartoon Animation, Preston Blair

Timing

Rate of acceleration conveys weight

Speed and acceleration of character’s movements convey emotion





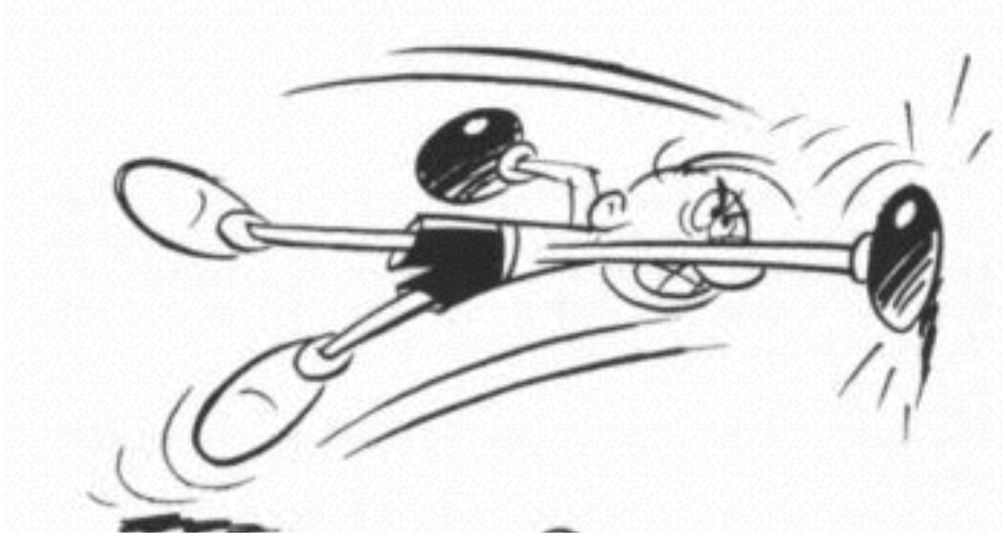
Timing for Animation, Whitaker & Halas

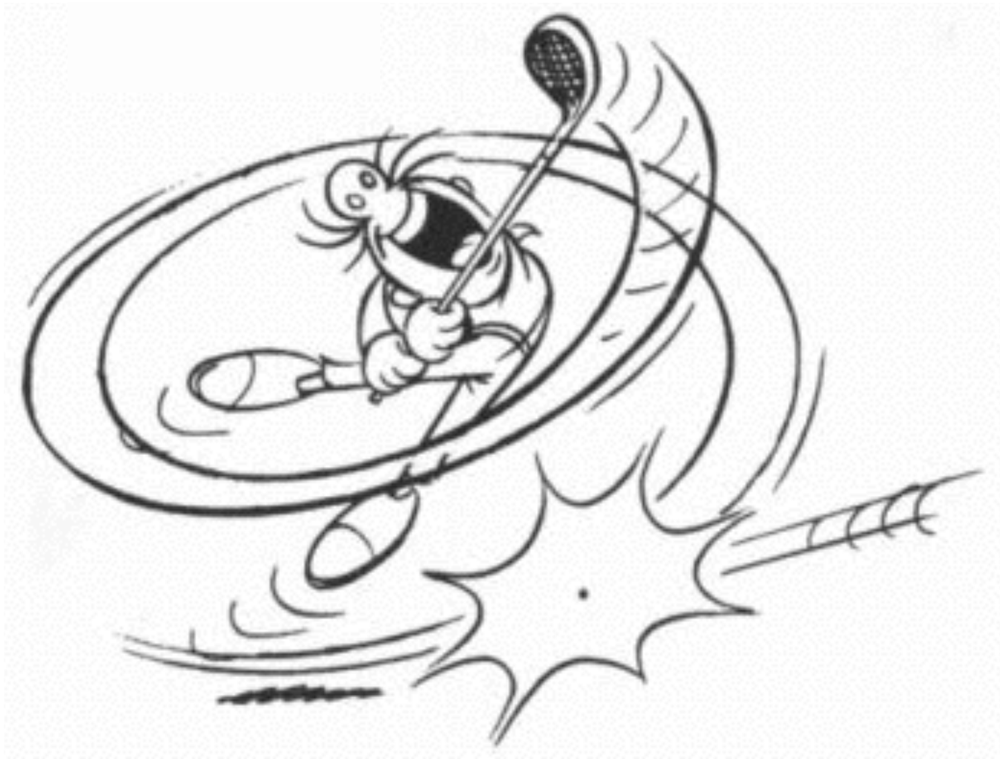
Exaggeration

Helps make actions clear

Helps emphasize story points and emotion

Must balance with non-exaggerated parts





Timing for Animation, Whitaker & Halas

Appeal

Attractive to the eye, strong design

Avoid symmetries

Disney Animation: The Illusion of Life

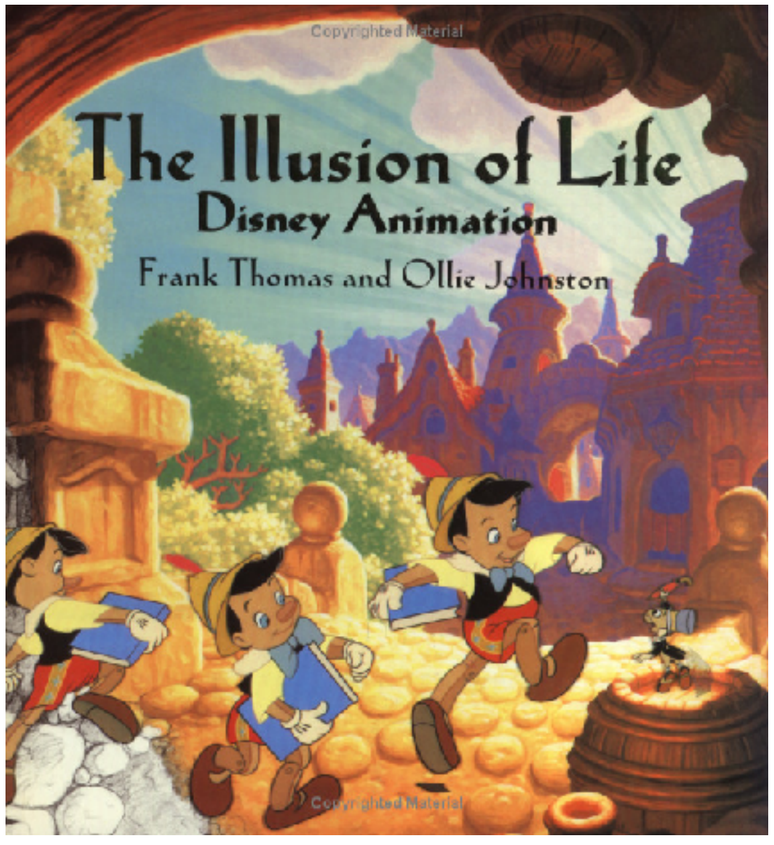
Personality

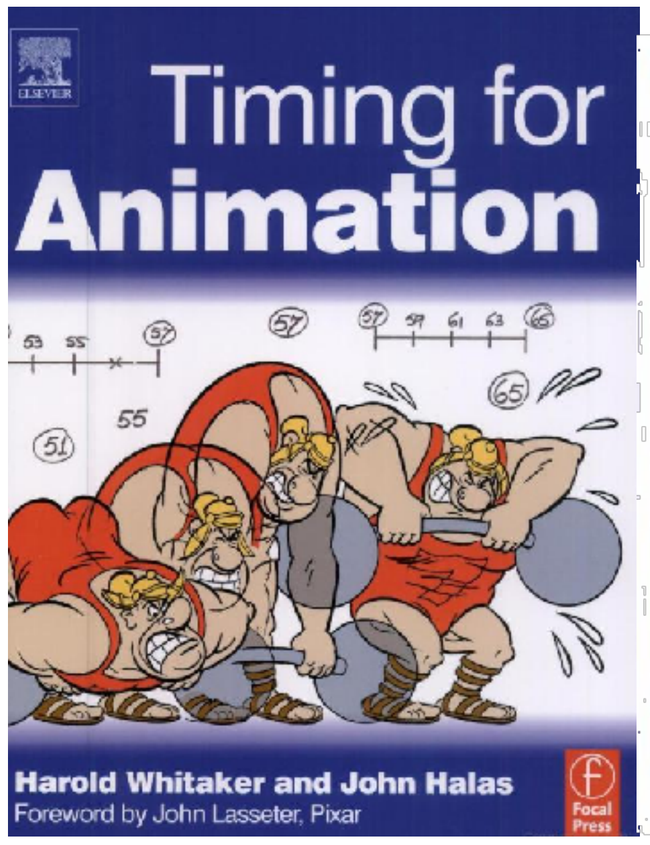
Action of character is result of its thoughts

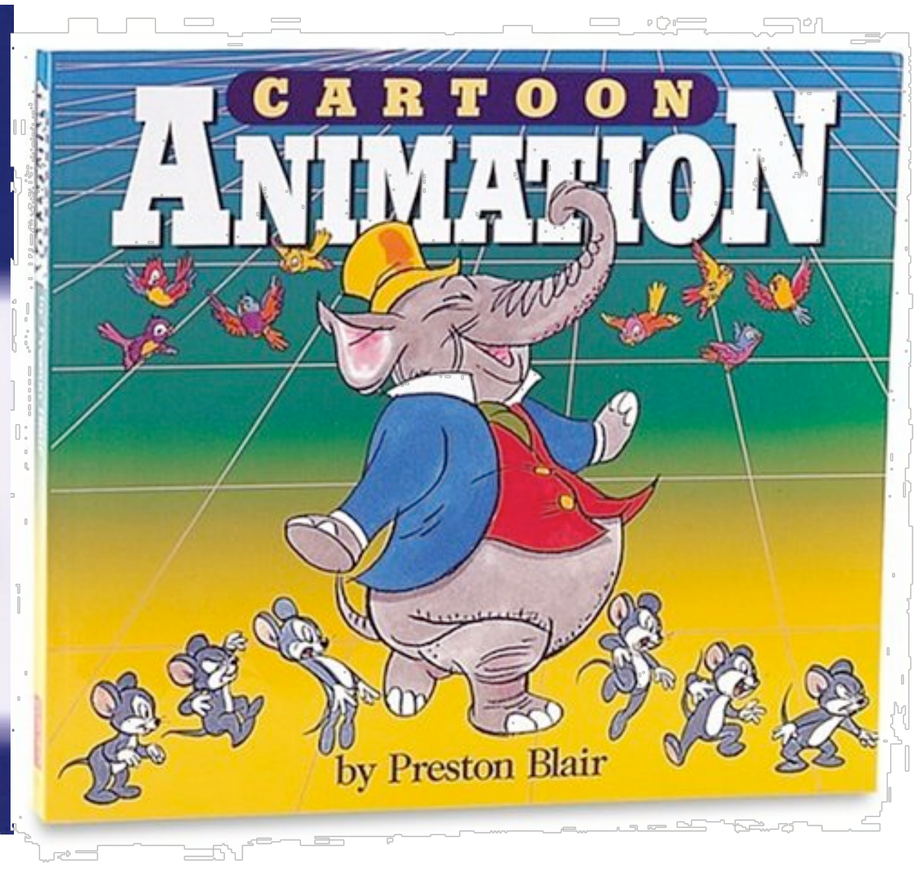
Know purpose & mood before animating each action

No two characters move the same way

Further Reading







12 Animation Principles

1. Squash and stretch
2. Anticipation
3. Staging
4. Straight ahead and pose-to-pose
5. Follow through
6. Ease-in and ease-out
7. Arcs
8. Secondary action
9. Timing
10. Exaggeration
11. Solid drawings
12. Appeal

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12 Animation Principles

Cento Lodgiani, https://vimeo.com/93206523

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12 Animation Principles

Applications:

- Movies
- Games
- User interfaces
- ...

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Computer Animation

Keyframe Animation

Keyframes

"Tweens"

Animator (e.g. lead animator) creates keyframes

Assistant (person or computer) creates in-between frames ("tweening")

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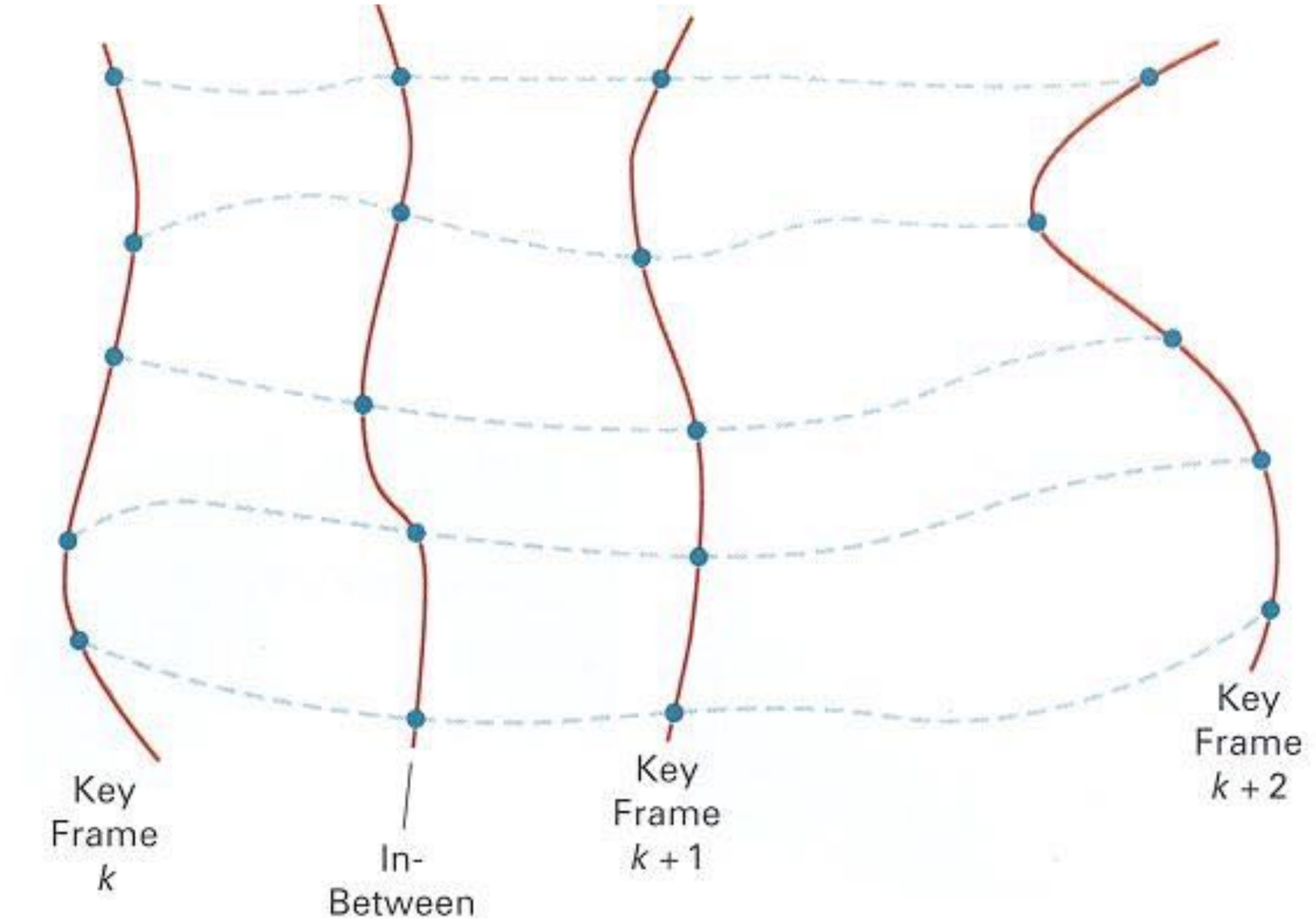
Keyframe Interpolation

Think of each frame as a vector of parameter values

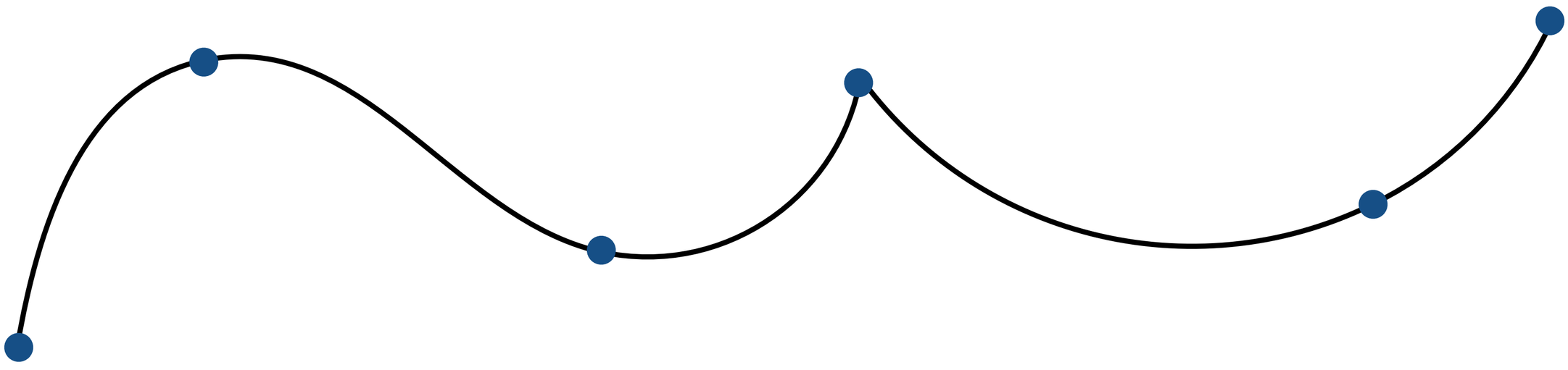


Hearn, Baker and Carithers, Figure 16.11

Keyframe Interpolation of Each Parameter

Linear interpolation usually not good enough

Recall splines for smooth / controllable interpolation



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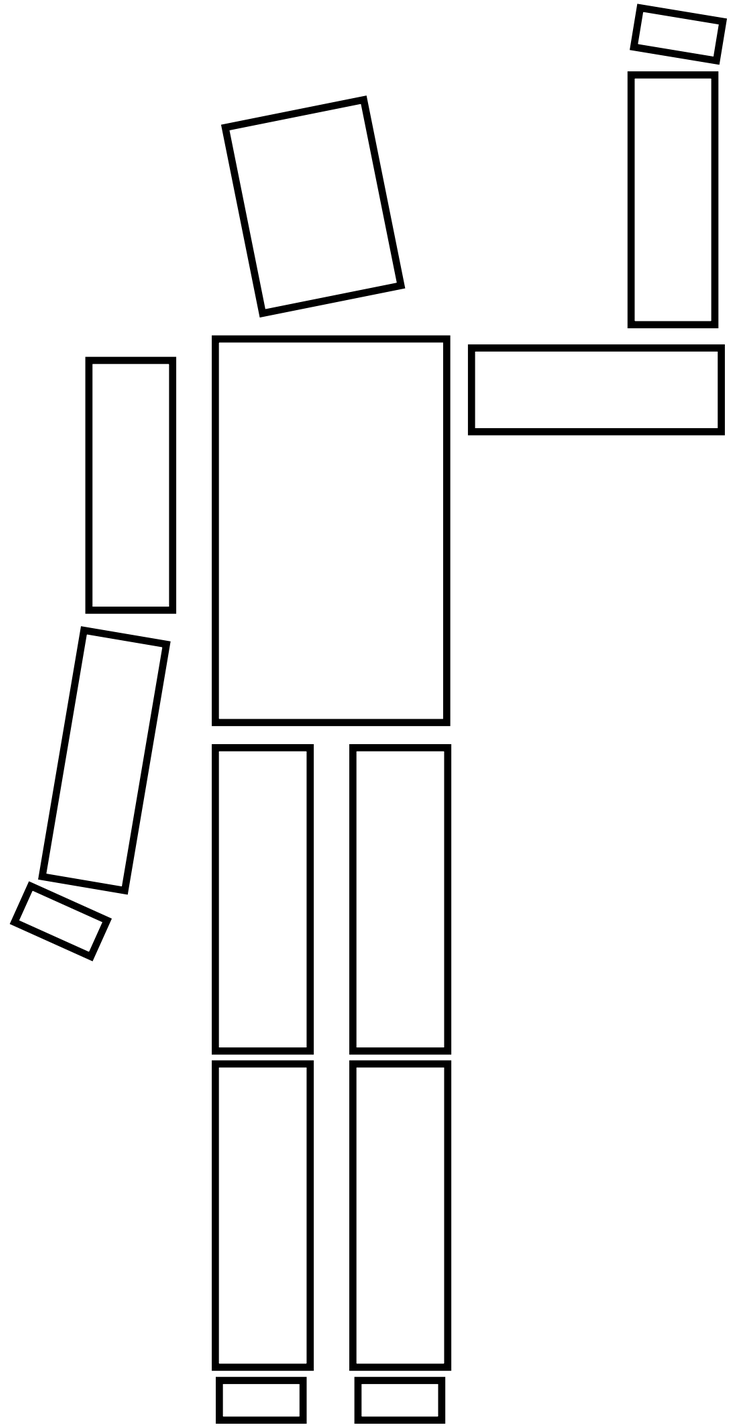
Forward Kinematics

Recall this skeleton from Transforms lecture

torso
- head
- right arm
  - upper arm
    - lower arm
      - hand
- left arm
  - upper arm
    - lower arm
      - hand

right leg
- upper leg
  - lower leg
    - foot

left leg
- upper leg
  - lower leg
    - foot



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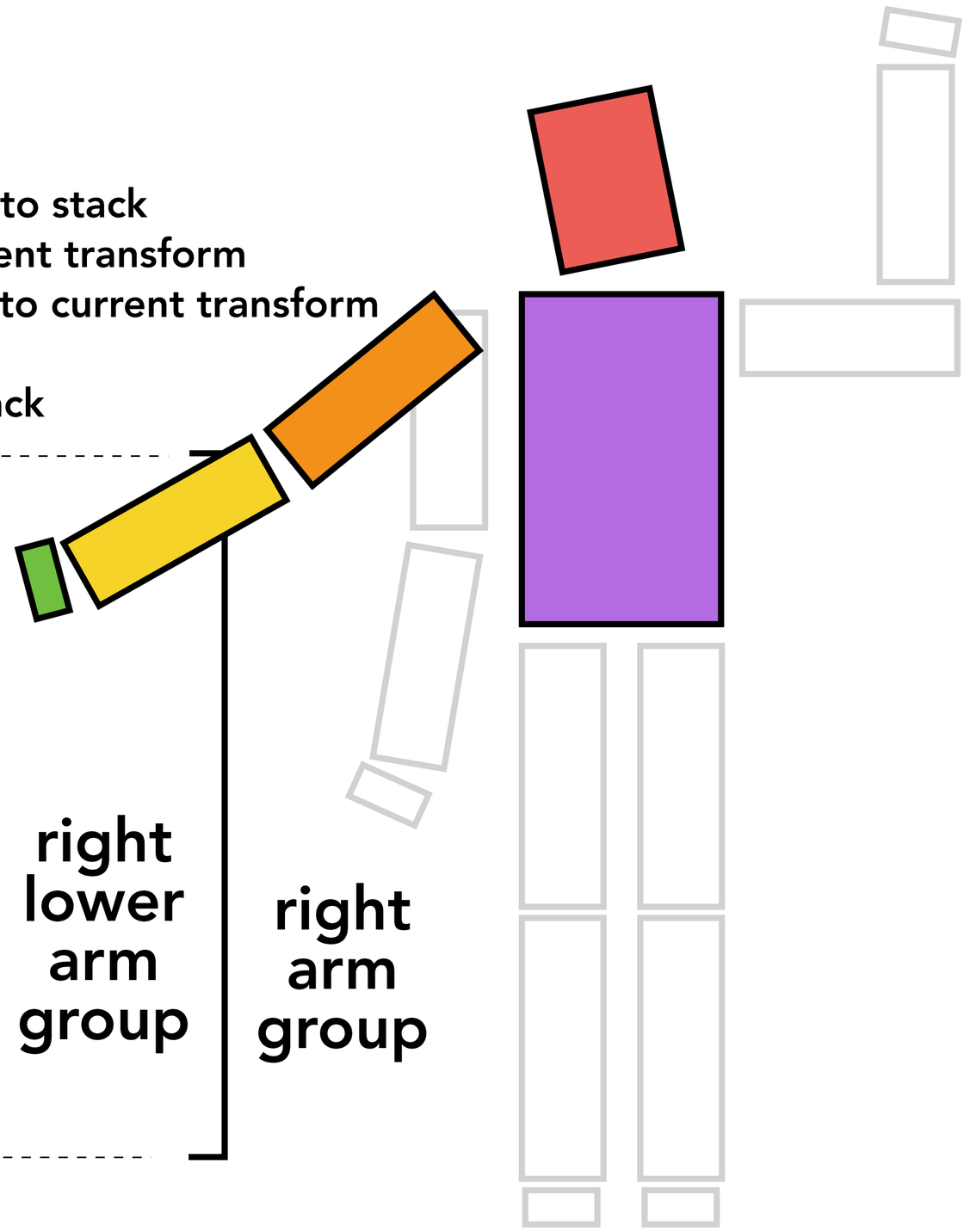
Skeleton - Hierarchical Representation

```
translate(0, 10);
    drawTorso();
        pushmatrix();  // push a copy of transform onto stack
            translate(0, 5);  // right-multiply onto current transform
            rotate(headRotation);  // right-multiply onto current transform
            drawHead();
        popmatrix();  // pop current transform off stack
        pushmatrix();
            translate(-2, 3);
        → rotate(rightShoulderRotation);
            drawUpperArm();
        pushmatrix();
            translate(0, -3);
            rotate(elbowRotation);
            drawLowerArm();
            pushmatrix();
                translate(0, -3);
                rotate(wristRotation);
                drawHand();
            popmatrix();
        popmatrix();
    popmatrix();
    ....
```

right hand

right lower arm group

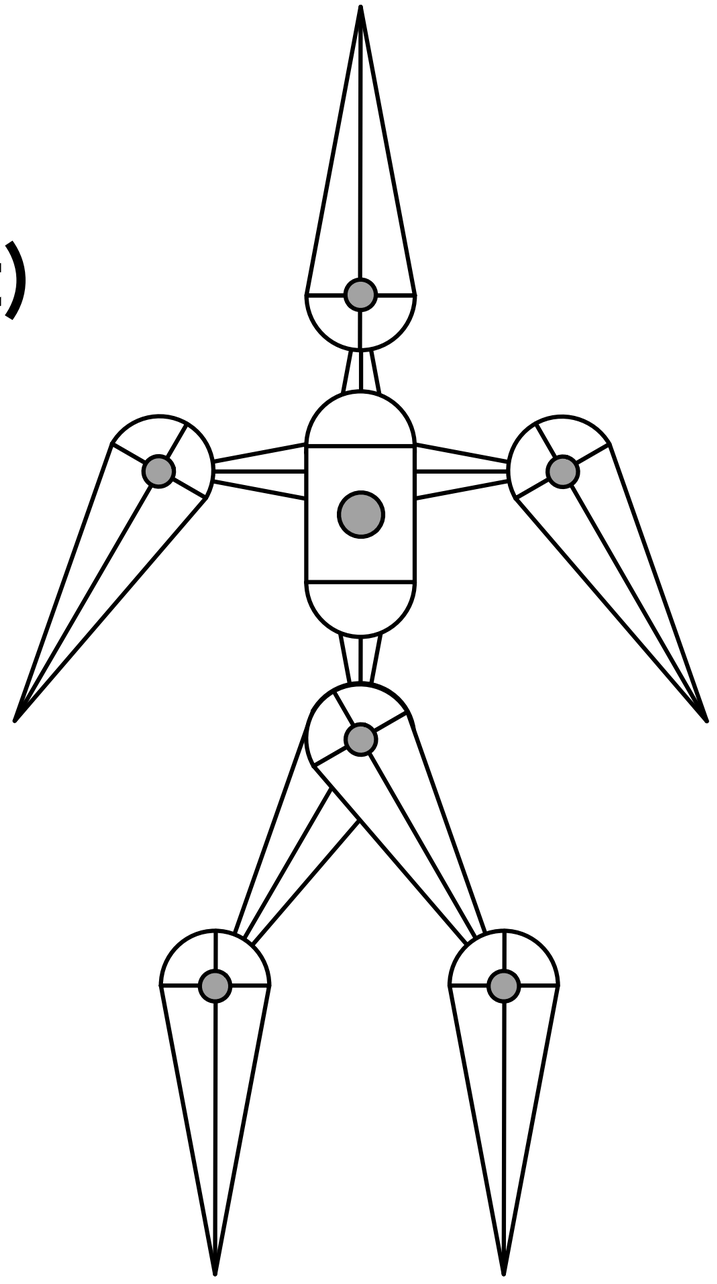
right arm group



Forward Kinematics

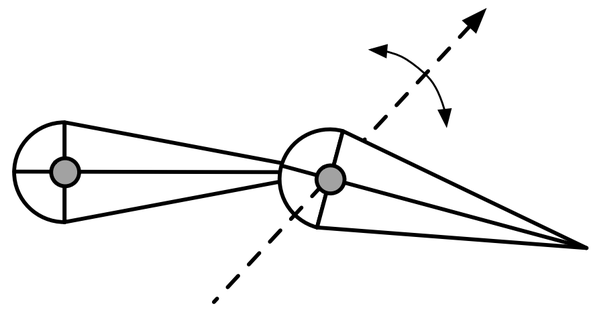
Articulated skeleton

- Topology (what's connected to what)
- Geometric relations from joints
- Tree structure (in absence of loops)

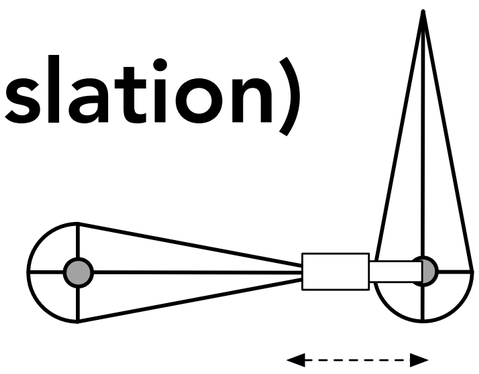


Joint types

- Pin (1D rotation)

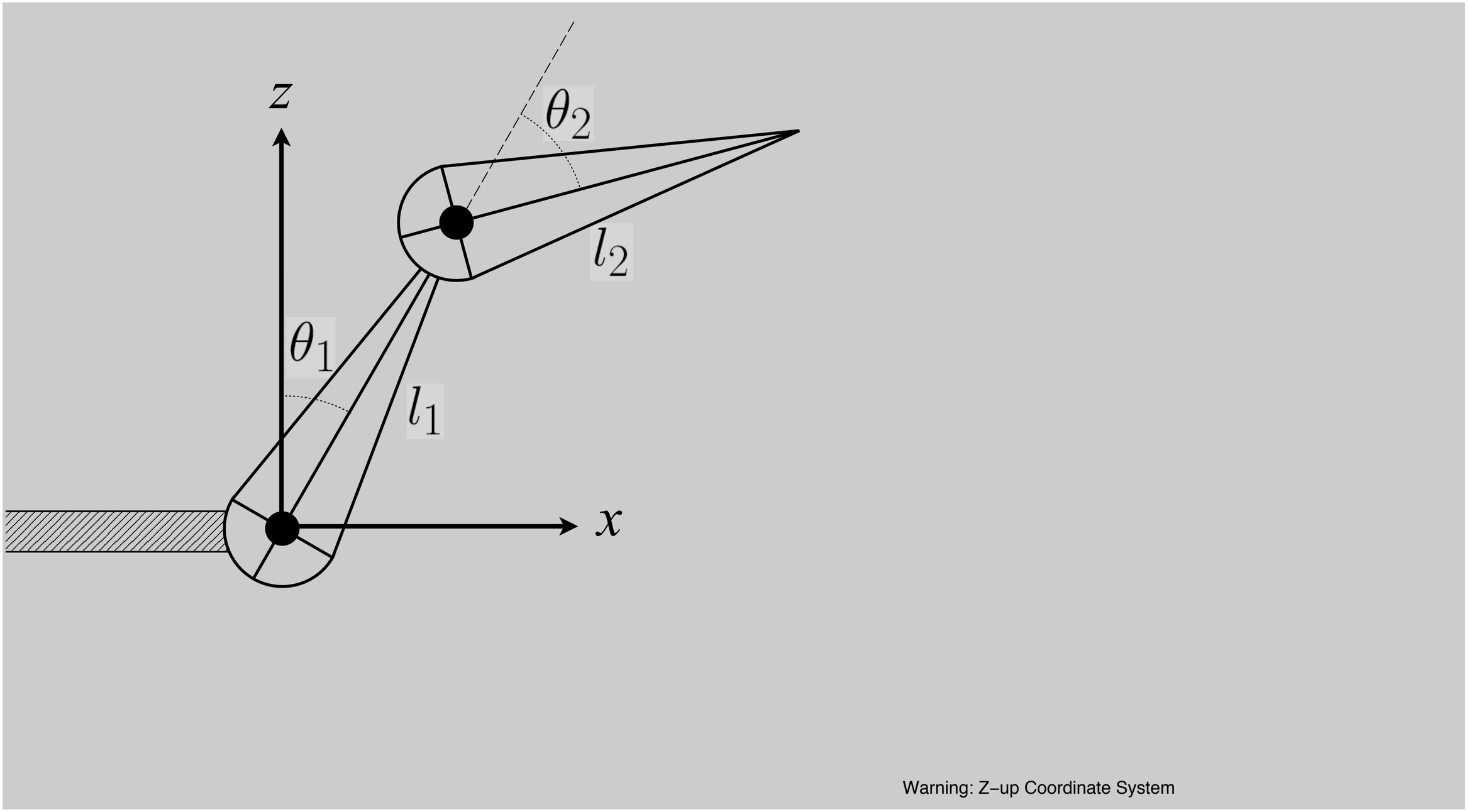


- Ball (2D rotation)
- Prismatic joint (translation)



Forward Kinematics

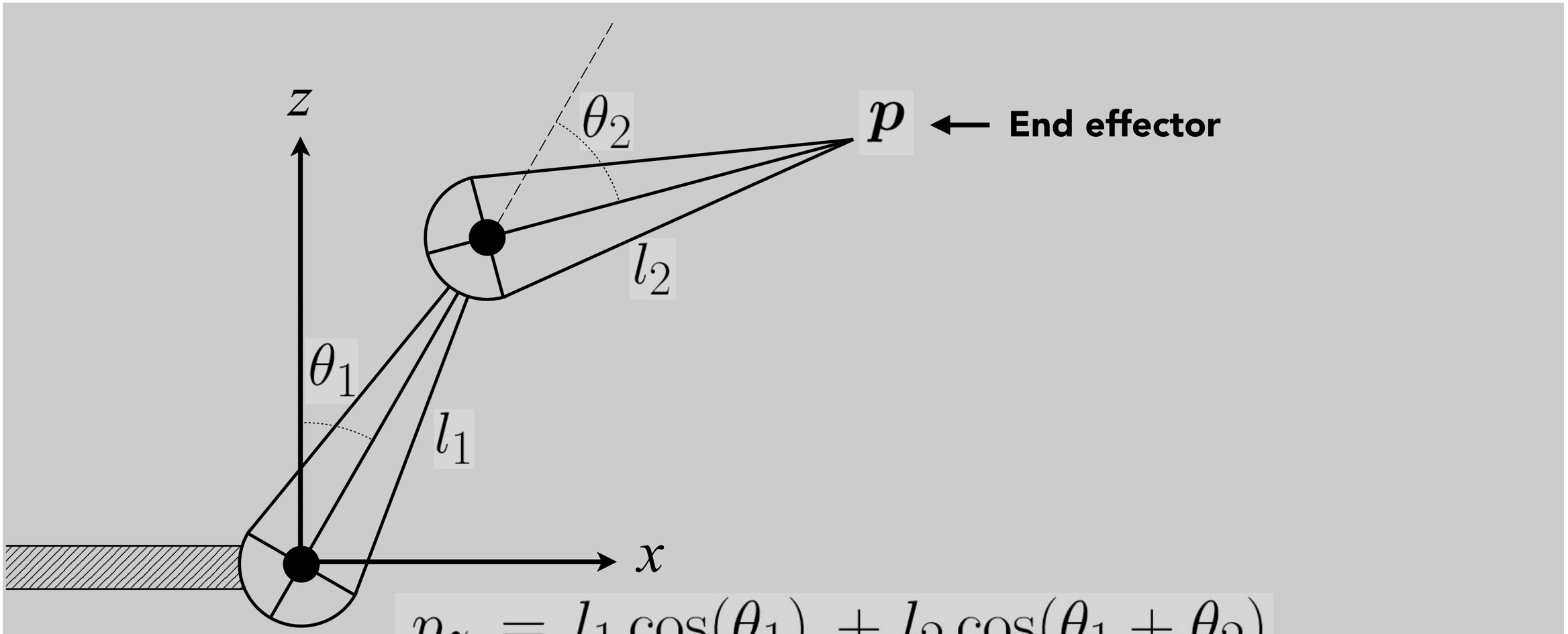
Example: simple two segment arm in 2D



Warning: Z–up Coordinate System

Forward Kinematics

Animator provides angles, and computer determines position *p* of end-effector



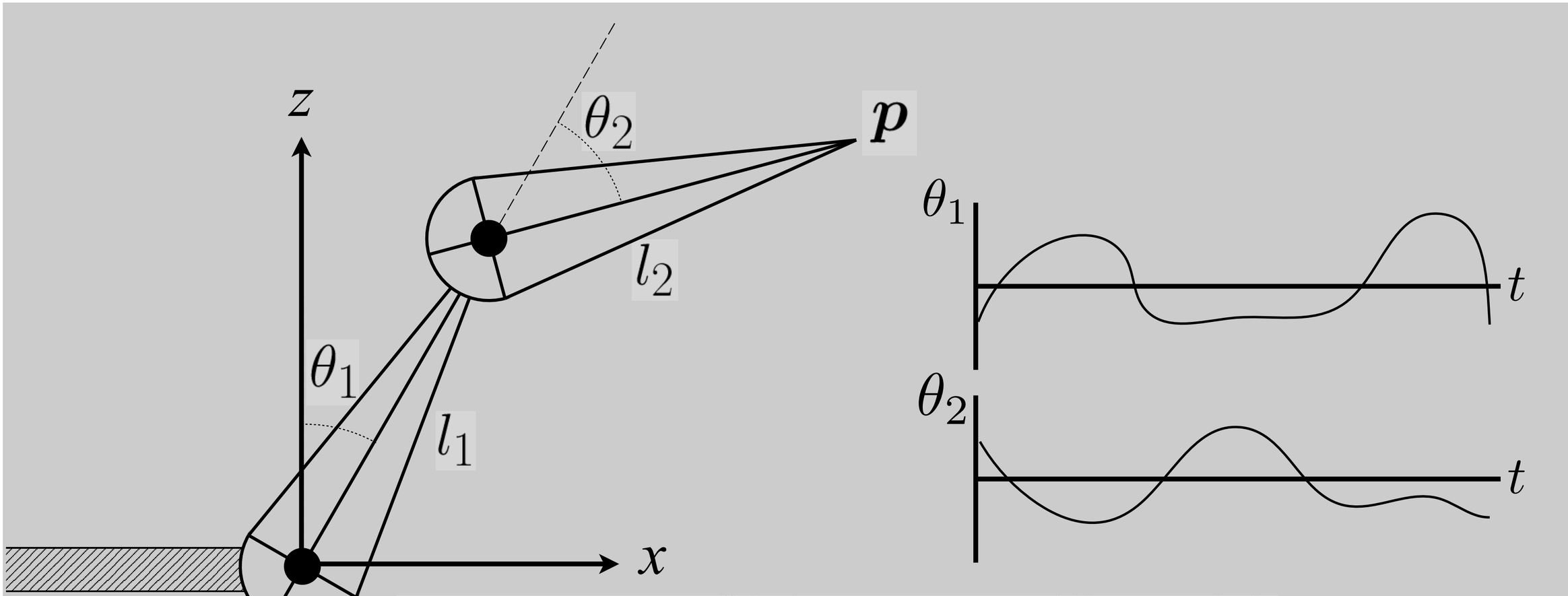
$$p_z = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2)$$

$$p_x = l_1 \sin(\theta_1) + l_2 \sin(\theta_1 + \theta_2)$$

Warning: Z–up Coordinate System

Forward Kinematics

Animation is described as angle parameter values as a function of time

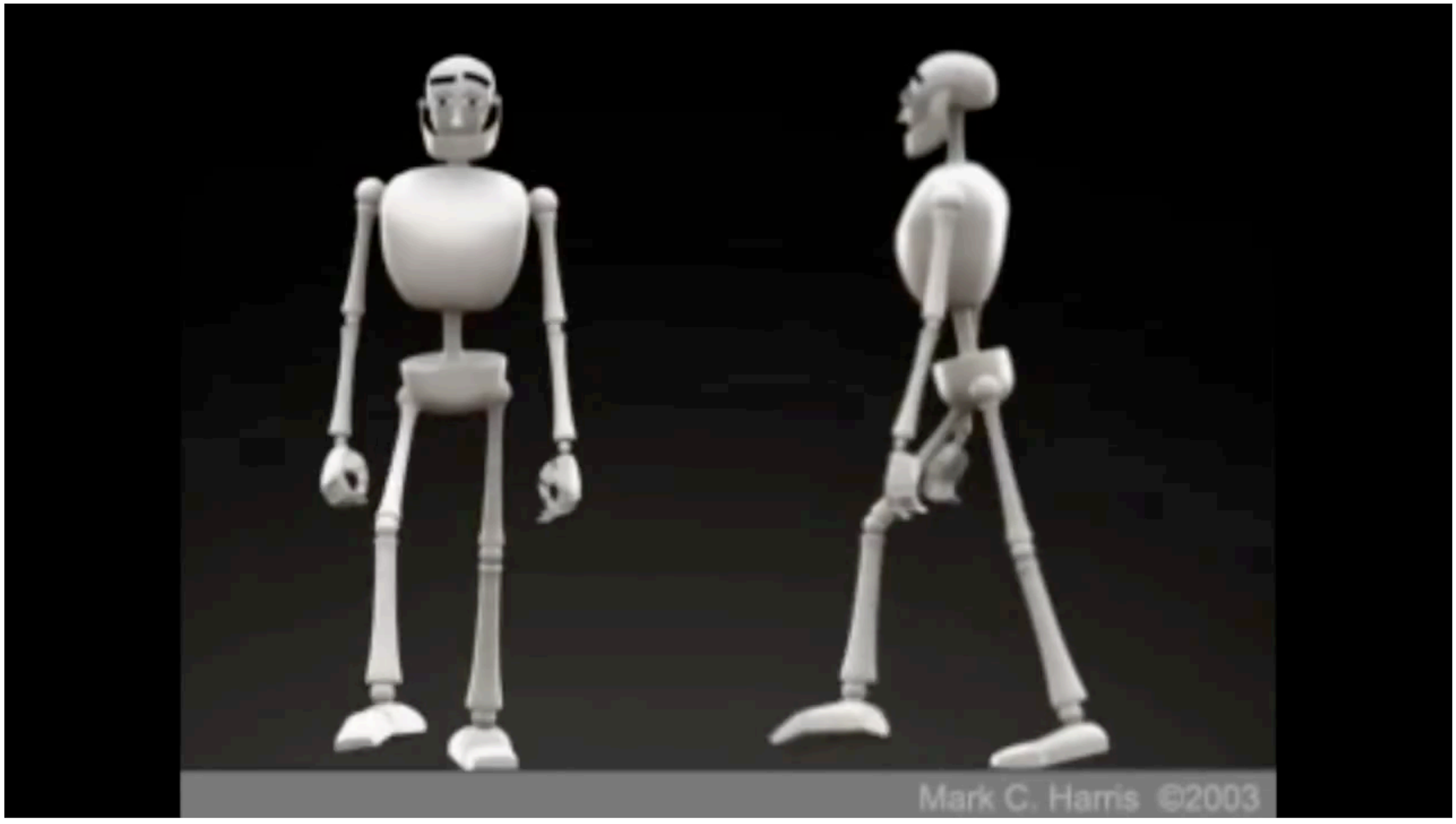


$$p_z = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2)$$

$$p_x = l_1 \sin(\theta_1) + l_2 \sin(\theta_1 + \theta_2)$$

Warning: Z–up Coordinate System

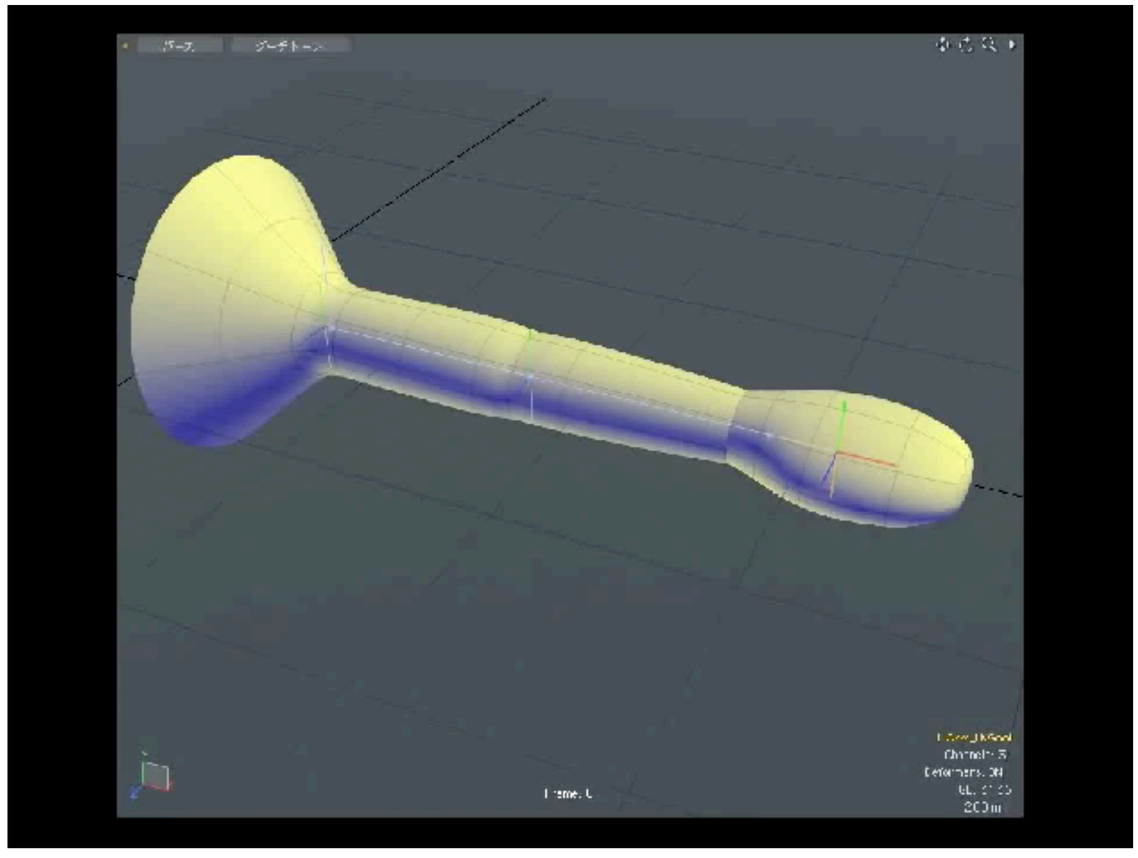
Example Walk Cycle



Inverse Kinematics

Inverse Kinematics

Given the end effector
position, find the joint angles.



Goals

- Keep end of limb fixed while body moves
- Position end of limb by direct manipulation
- (More general: arbitrary constraints)

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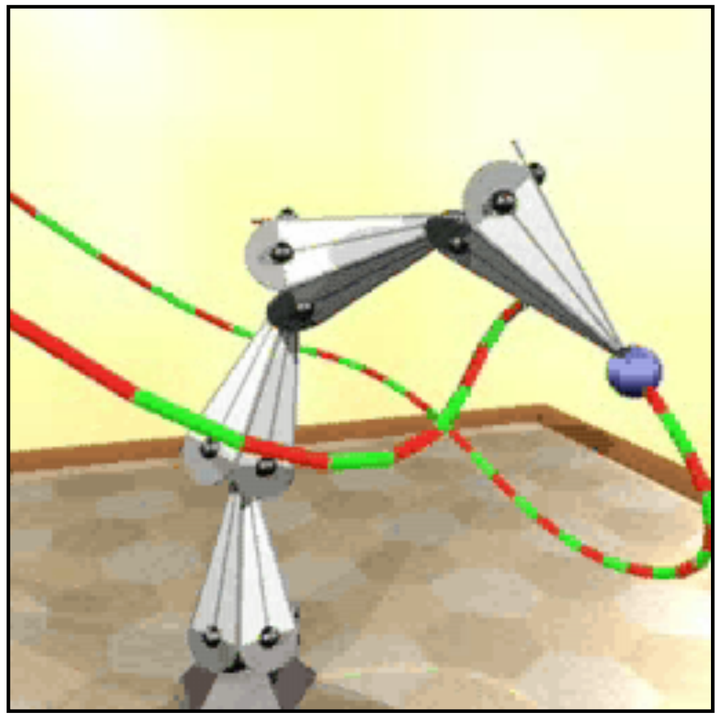
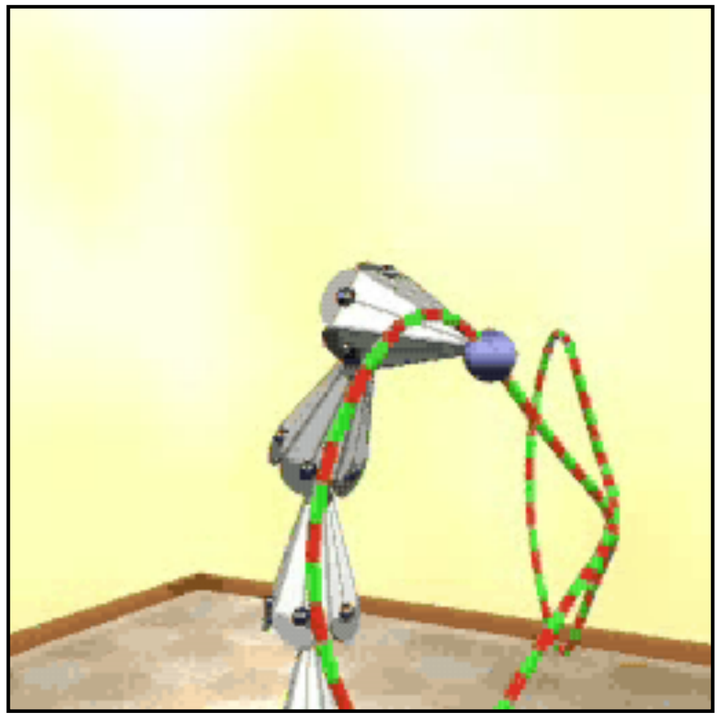
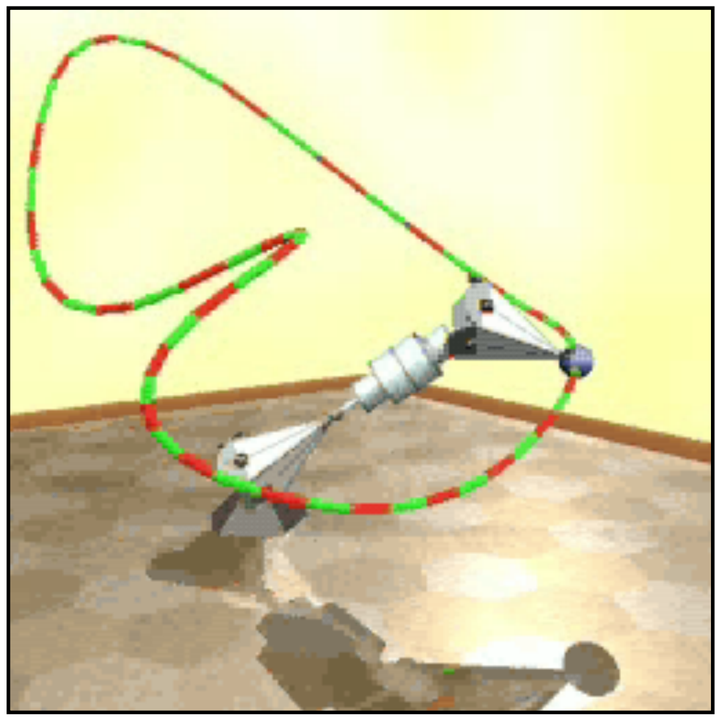
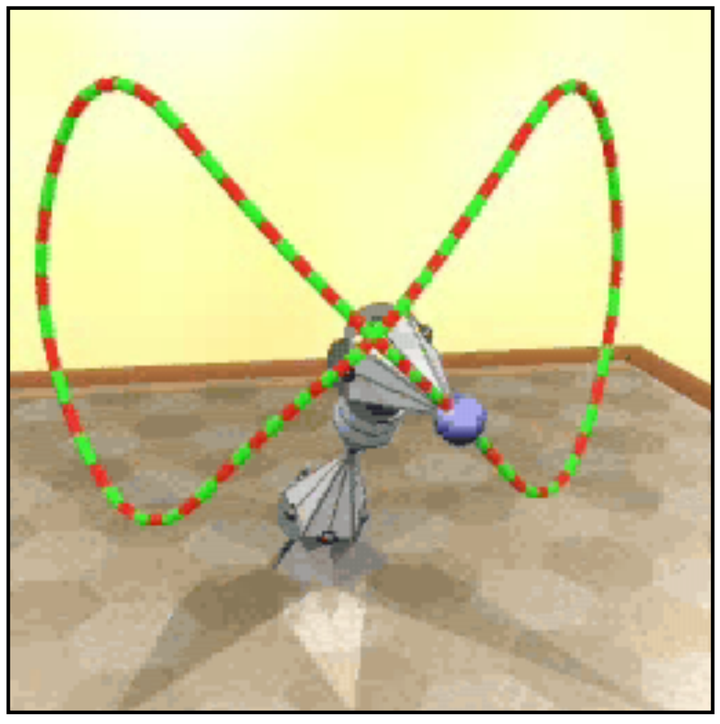
Inverse Kinematics



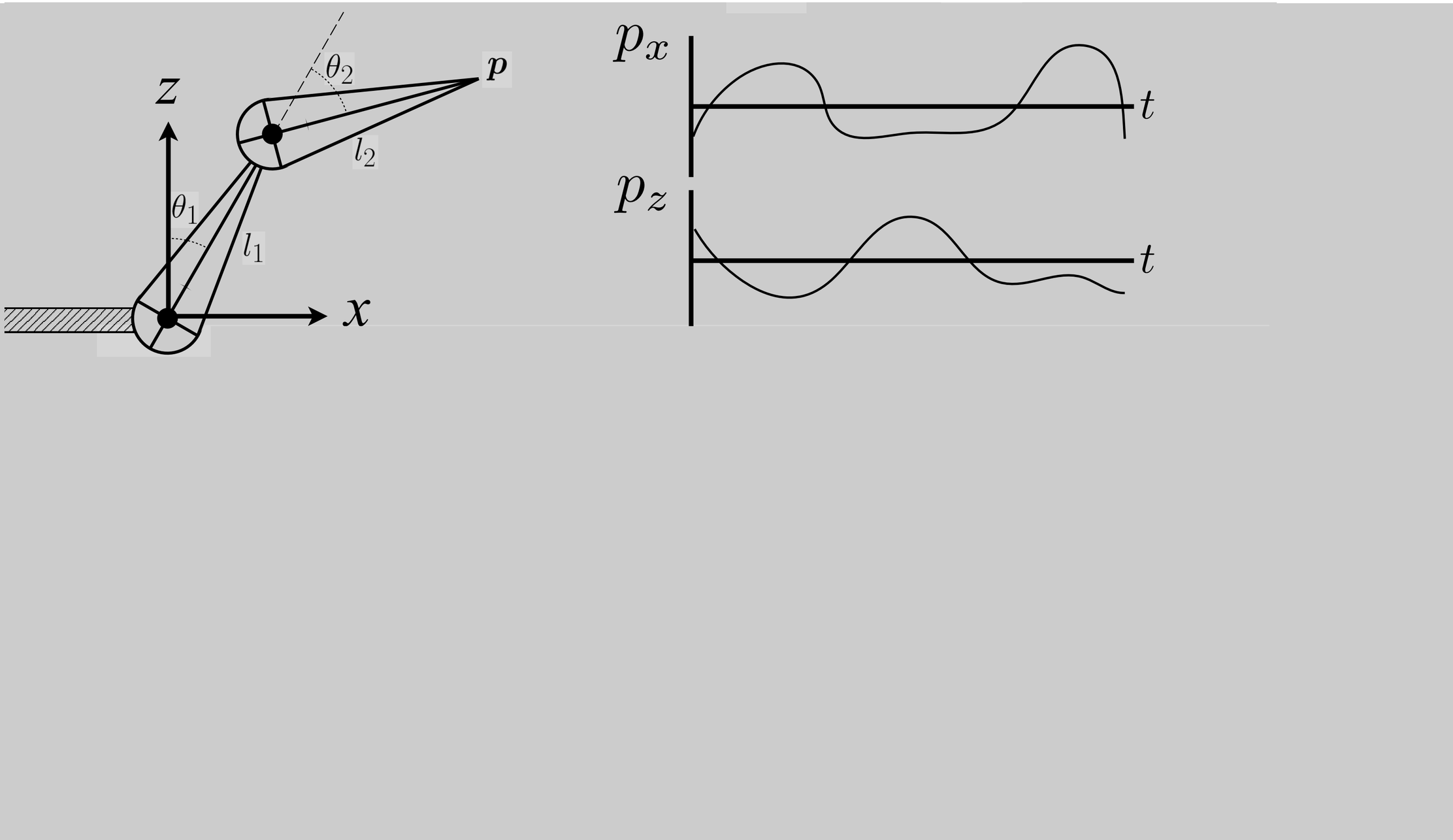






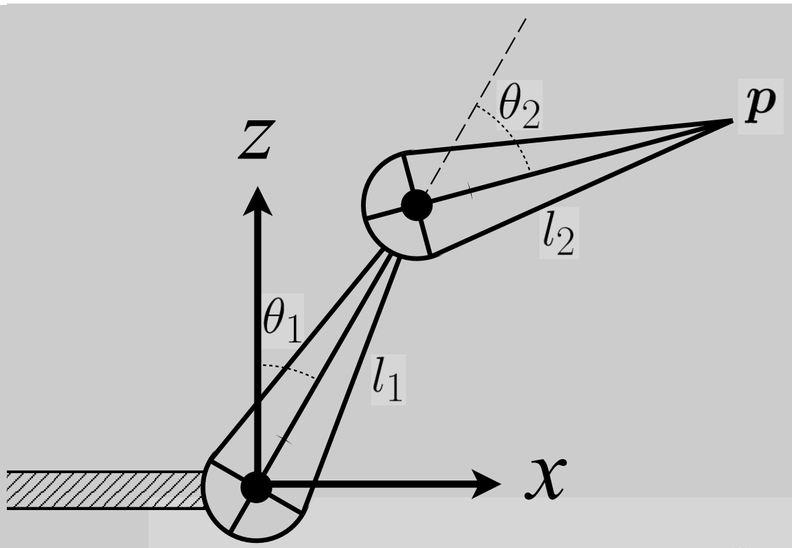
Inverse Kinematics

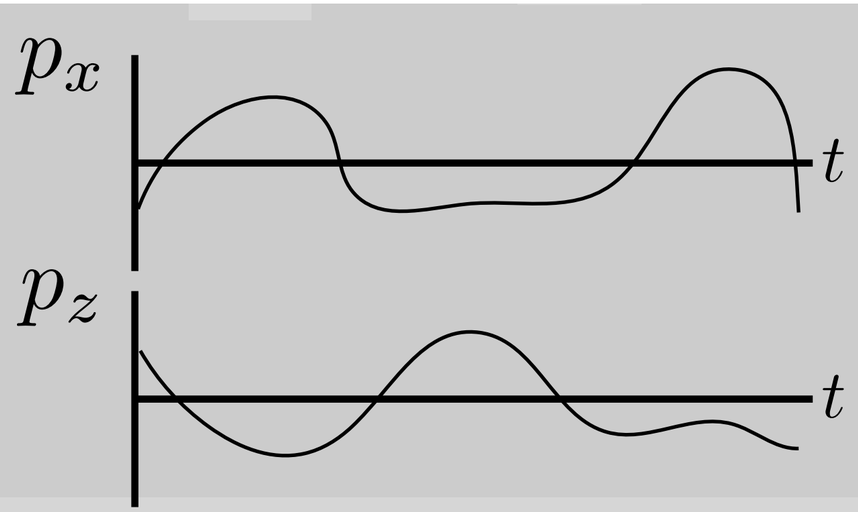
Animator provides position of end-effector, and computer must determine joint angles that satisfy constraints



Inverse Kinematics

Direct inverse kinematics: for two-segment arm, can solve for parameters analytically





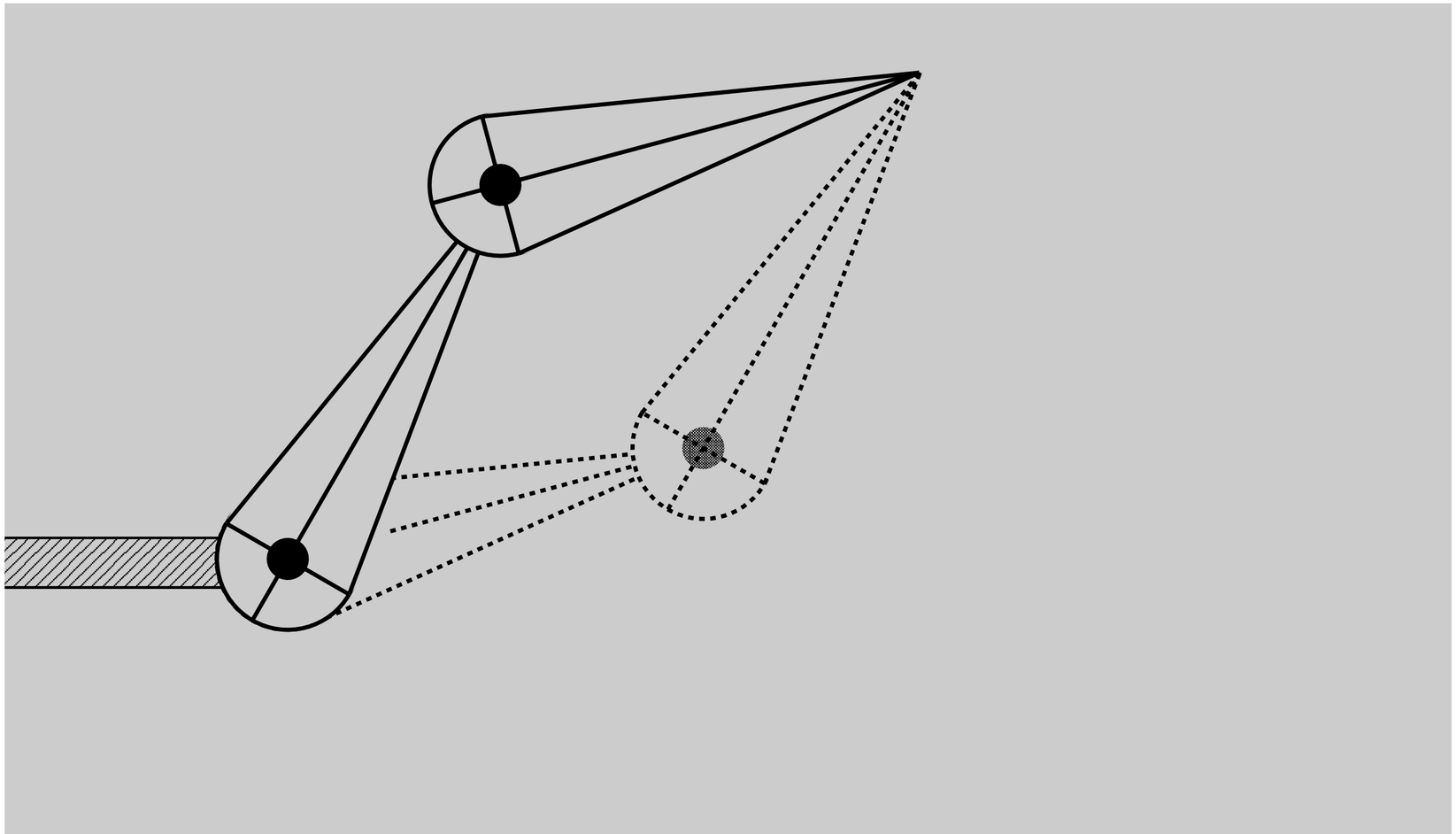
$$\theta_2 = \cos^{-1}\left(\frac{p_z^2 + p_x^2 - l_1^2 - l_2^2}{2 l_1 l_2}\right)$$

$$\theta_1 = \frac{-p_z l_2 \sin(\theta_2) + p_x(l_1 + l_2\cos(\theta_2))}{p_x l_2 \sin(\theta_2) + p_z(l_1 + l_2\cos(\theta_2))}$$

Inverse Kinematics

Why is the problem hard?

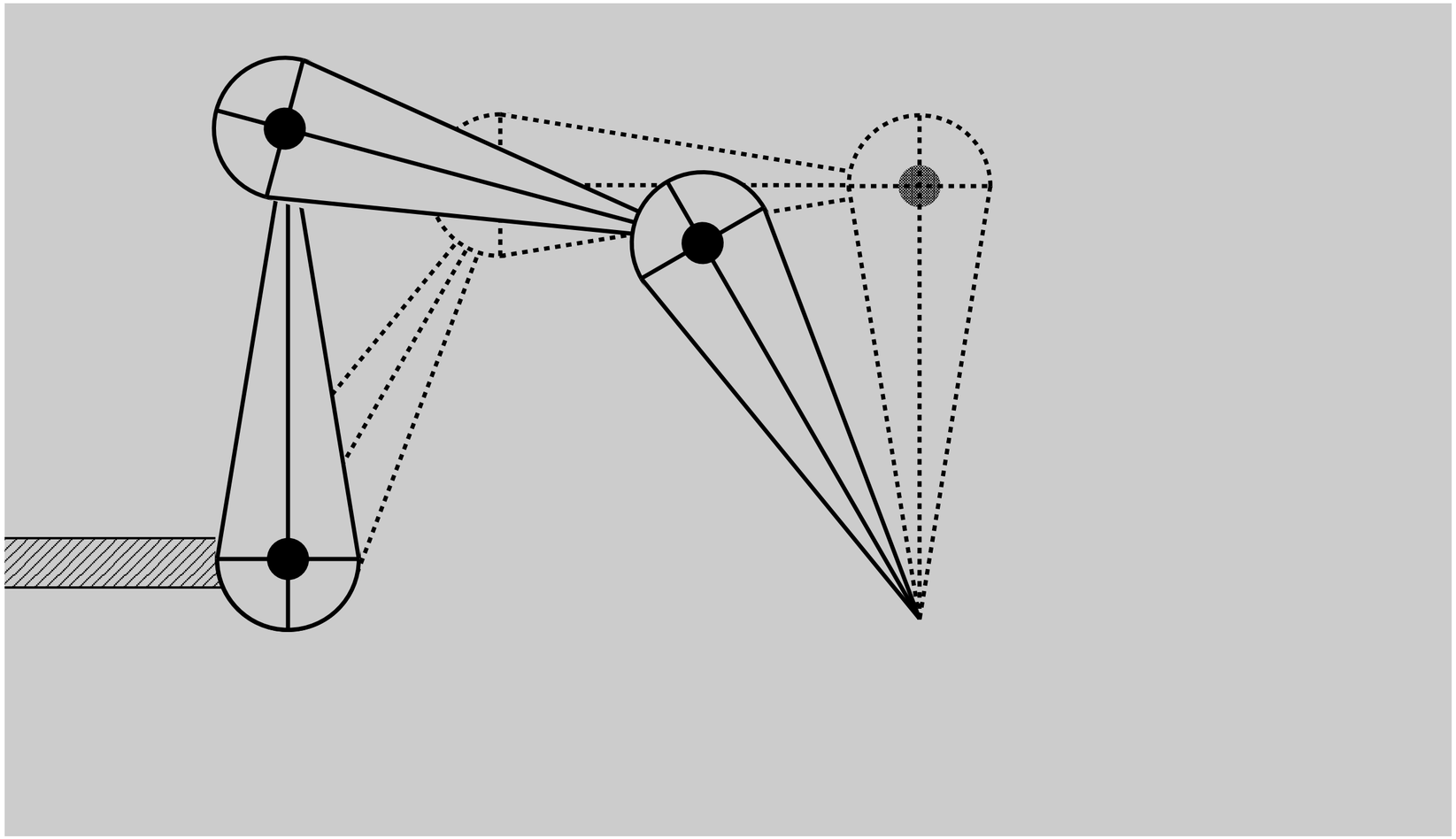
- Multiple solutions separated in configuration space



Inverse Kinematics

Why is the problem hard?

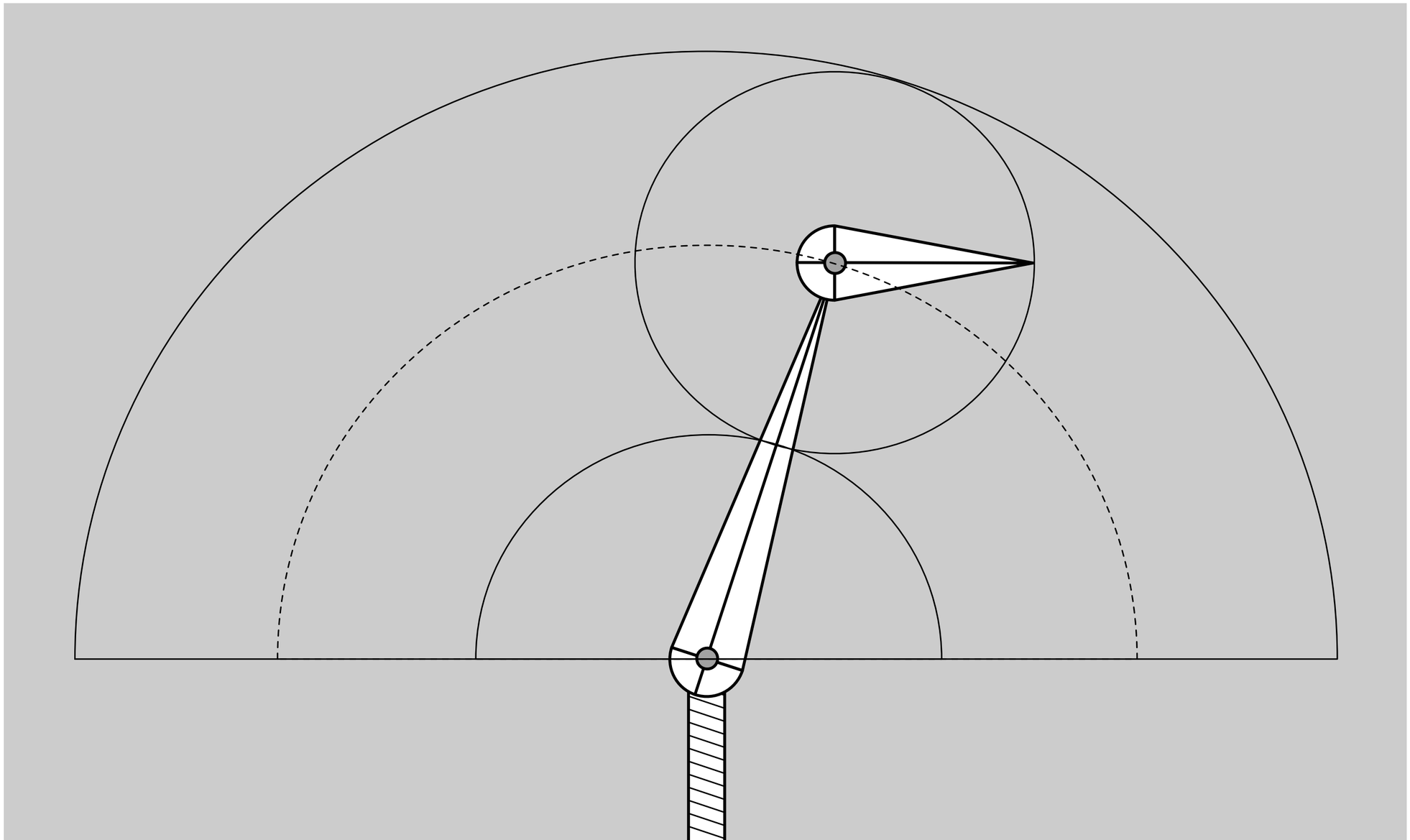
- Multiple solutions connected in configuration space



Inverse Kinematics

Why is the problem hard?

- Solutions may not always exist

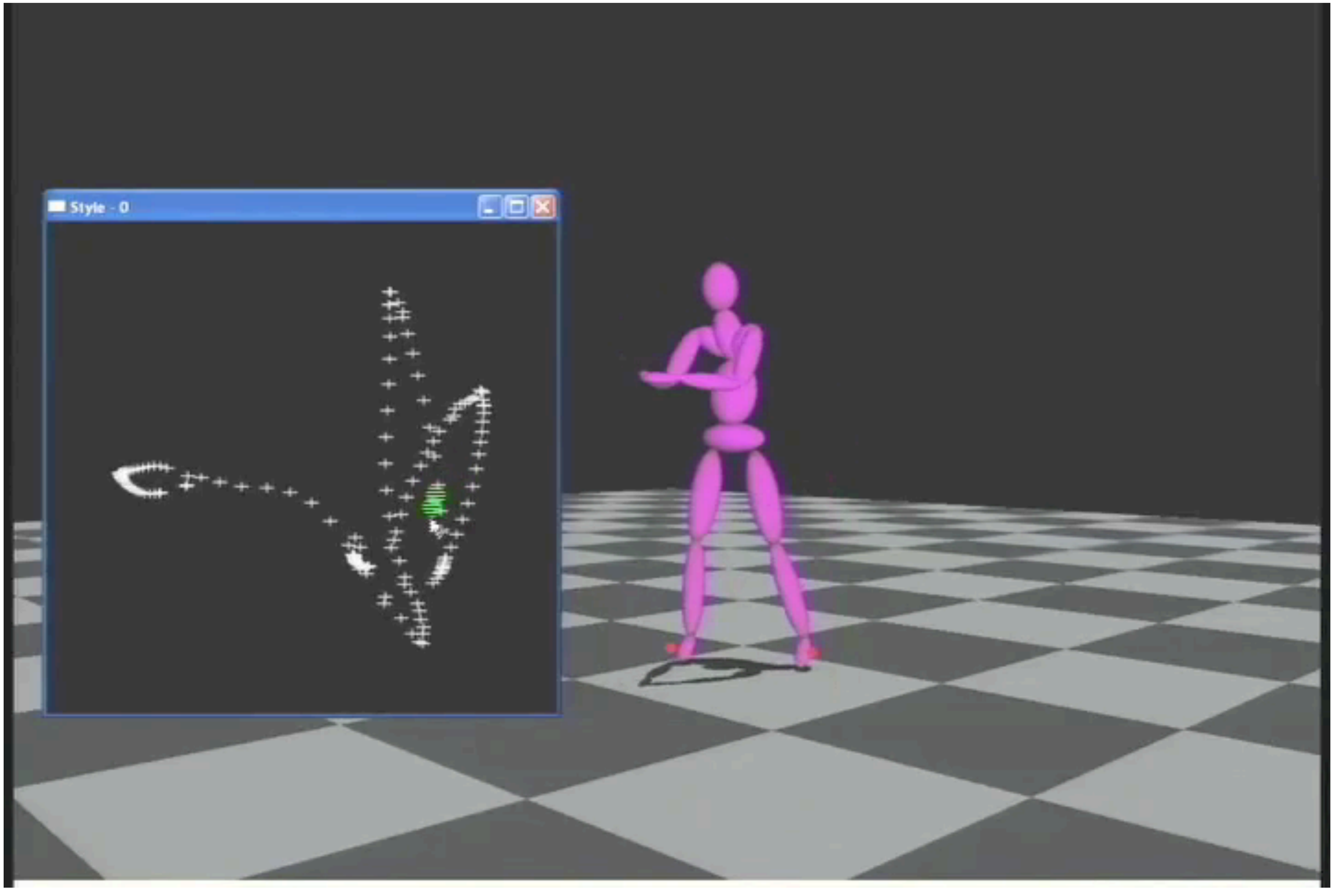


Inverse Kinematics

Numerical solution to general N-link IK problem

- Choose an initial configuration
- Define an error metric (e.g. square of distance between goal and current position)
- Compute gradient of error as function of configuration
- Apply gradient descent (or Newton's method, or other optimization procedure)

Style-Based IK



Grochow et al., Style Based Inverse Kinematics

Kinematics Pros and Cons

Strengths

- Direct control is convenient
- Implementation is straightforward

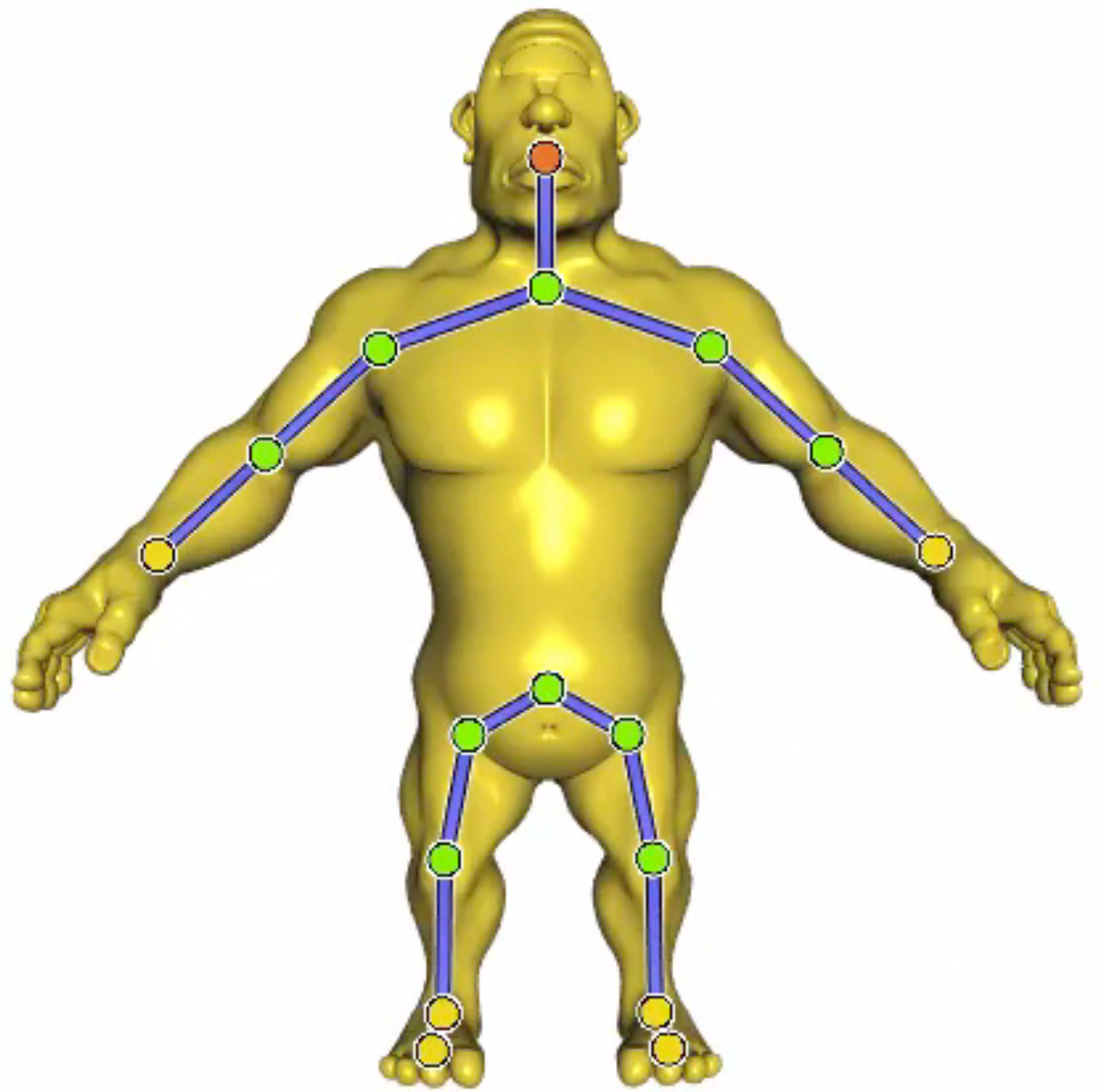
Weaknesses

- Animation may be inconsistent with physics
- Time consuming for artists

Skinning

Skinning

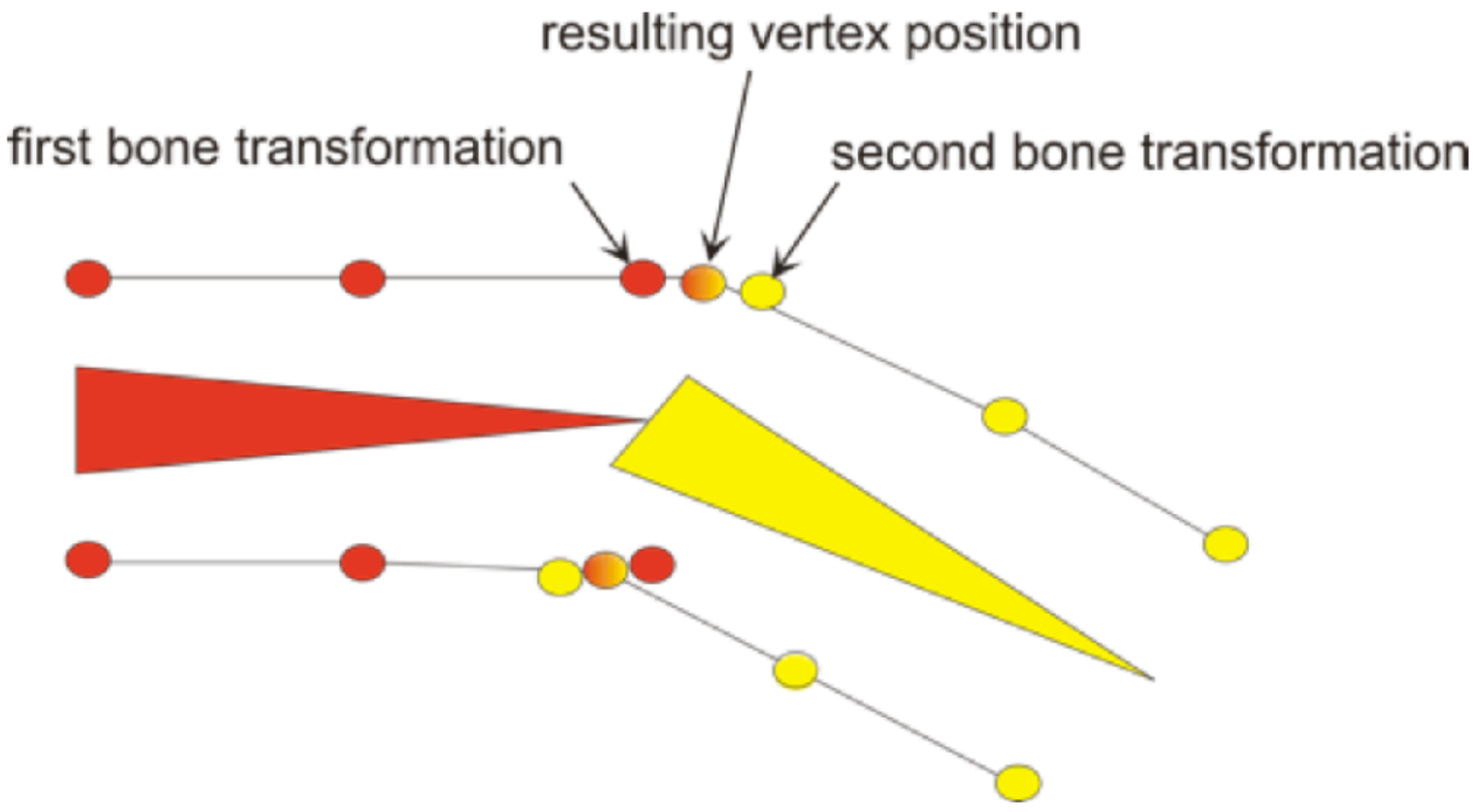
Goal: move the surface along with assigned bones or "handles"



Courtesy of Alec Jacobson

Basic Idea

1. Transform each vertex with each bone rigidly
2. Blend the results using weights, or assignments



Common Approach: Linear Blend Skinning (LBS)

Blend contribution linearly.

Super simple to implement. Great for real time.

How much influence this bone has on v (often sparse)

Bone j transformation

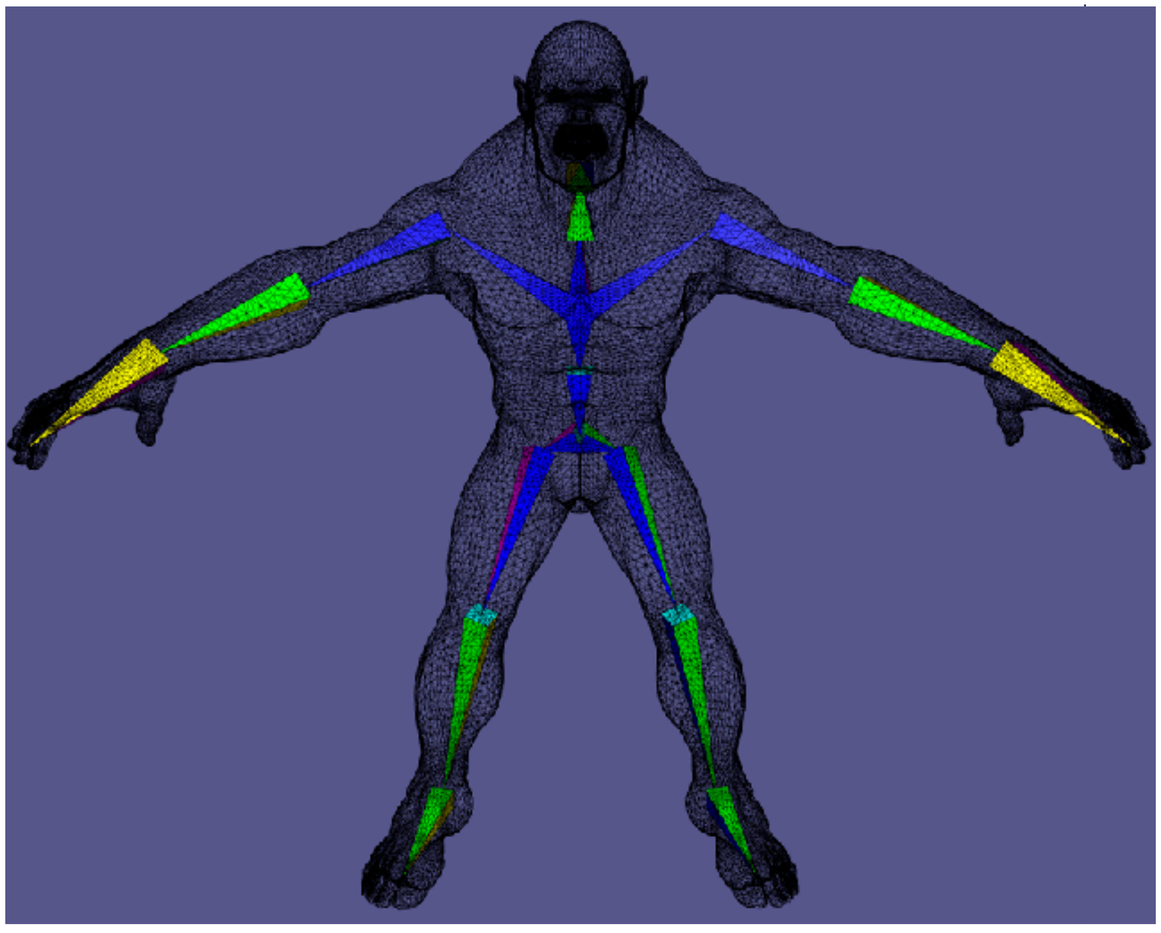
$$\mathbf{v}' = \sum_{j \in H} w_j(\mathbf{v}) \mathbf{T}_j \begin{pmatrix} \mathbf{v} \\ 1 \end{pmatrix}$$

New vertex

Original vertex

Ren Ng
Presented by David McAllister

Illustration of Rig & Skinning Weights



Bone transformations

Skinning Weights

CS184

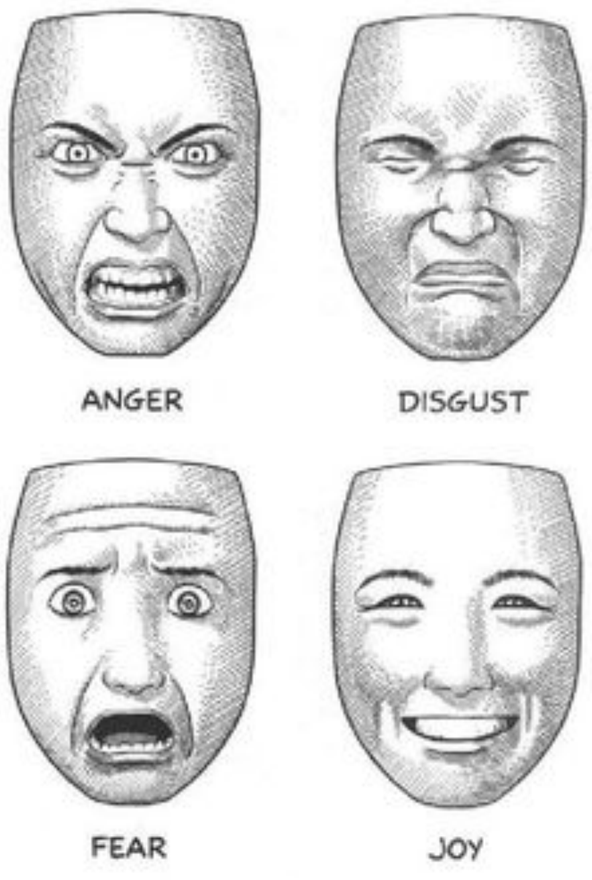
Ren Ng

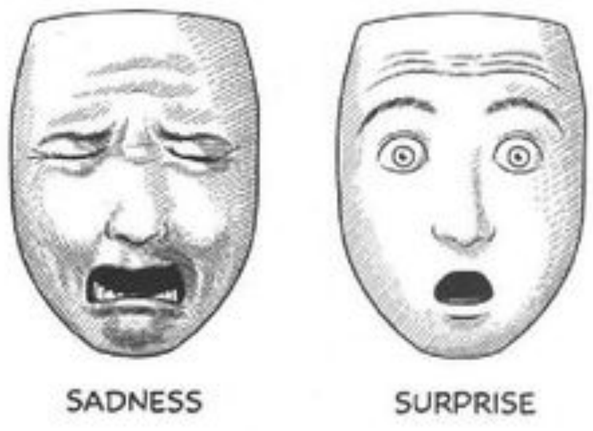
Presented by David McAllister

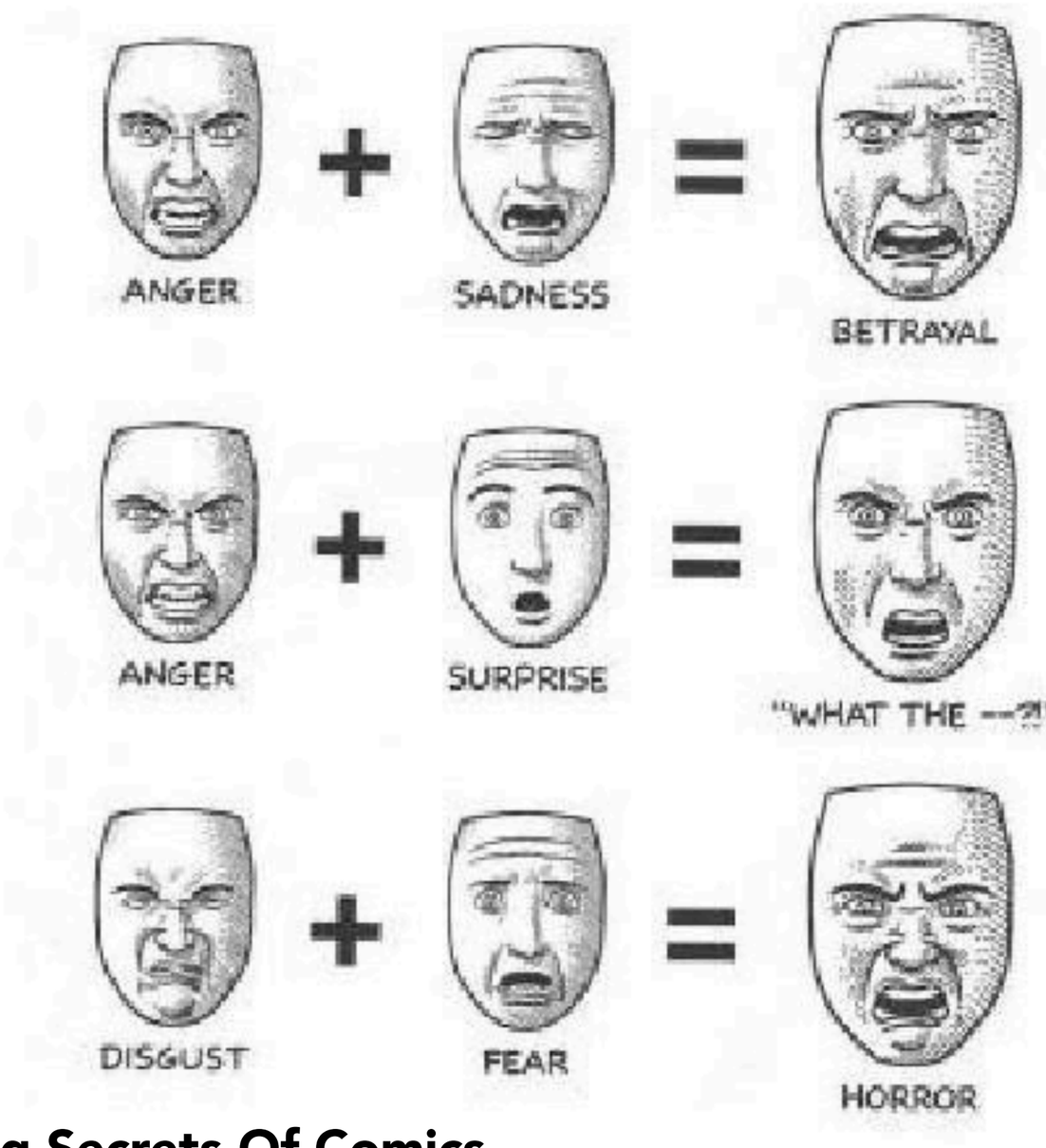
Blend Shapes

Blend Shapes

Not all deformation is from bones.
Interpolate surfaces between key shapes







CS184

Making Comics: Storytelling Secrets Of Comics, Manga, and Graphics Novels by Scott McCloud

Ren Ng

Presented by David McAllister

Blend Shapes

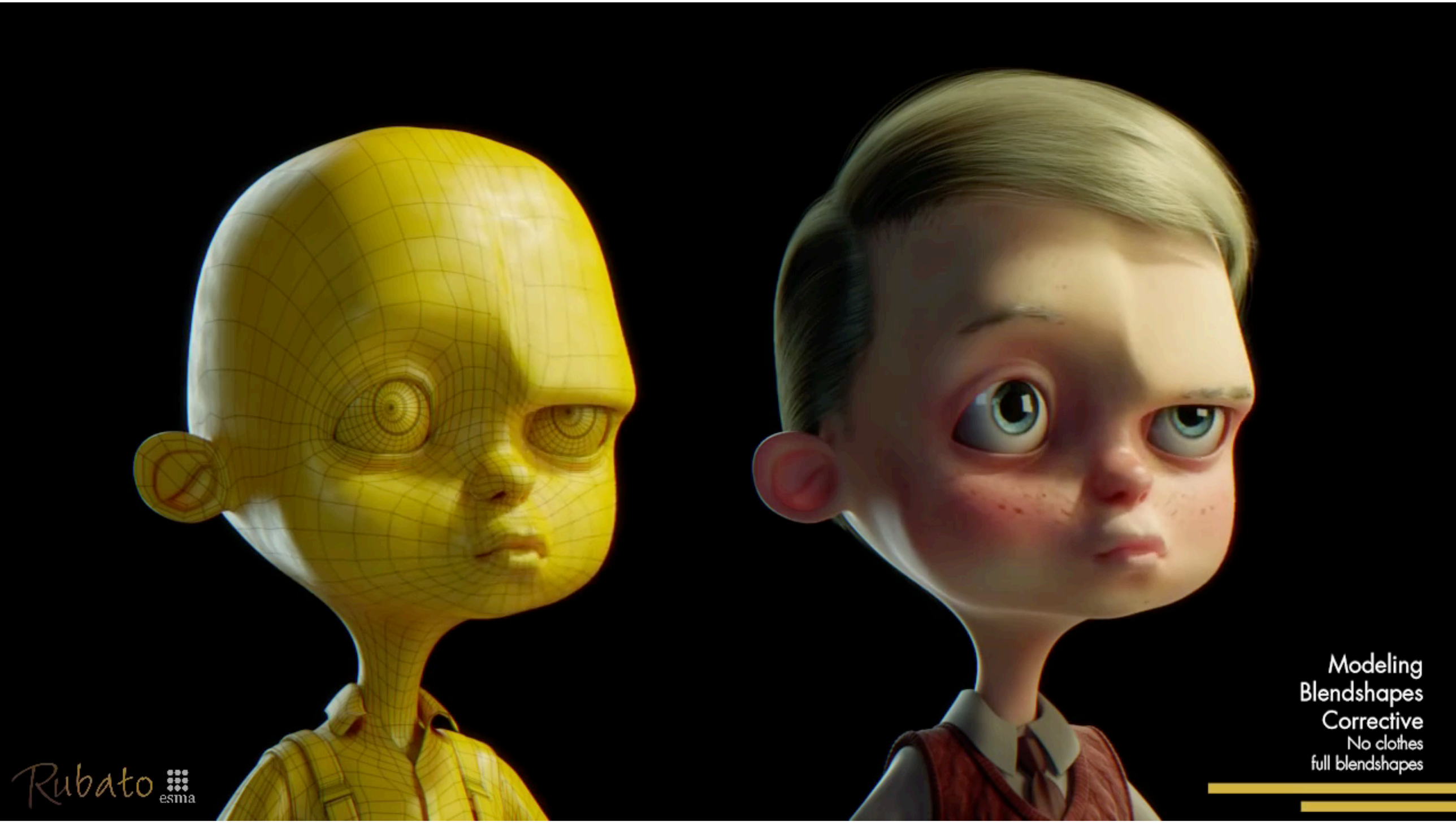
- A set of vertex offsets to neutral shape
- Linearly interpolate these key blend shapes for control
- Often used for expressions
- Works for deformations that are linear,
  i.e. the average of two shapes is a valid shape

$$B = \mathrm{vec}\left(\begin{bmatrix} \Delta x_1 & \Delta y_1 & \Delta z_1 \\ & \vdots & \\ \Delta x_N & \Delta y_N & \Delta z_N \end{bmatrix}\right)$$

$$V = \sum_i \beta_i B_i$$

Ren Ng
Presented by David McAllister

Blend Shapes



Courtesy Félix Ferrand

Rigging

Rigging

Augment character with controls to
easily change its pose, create facial expressions, bulge muscles, etc.

Rigging is like the strings on a marionette.

Capture space of meaningful deformations.

Varies from character to character.

Skeleton is ONE type of rigging

Ren Ng
Presented by David McAllister

Example of A Diverse Set of Sophisticated Rigs

CS184 **Courtesy of Matthew Lailler** **Ren Ng**

Presented by David McAllister

Motion Capture

Motion Capture

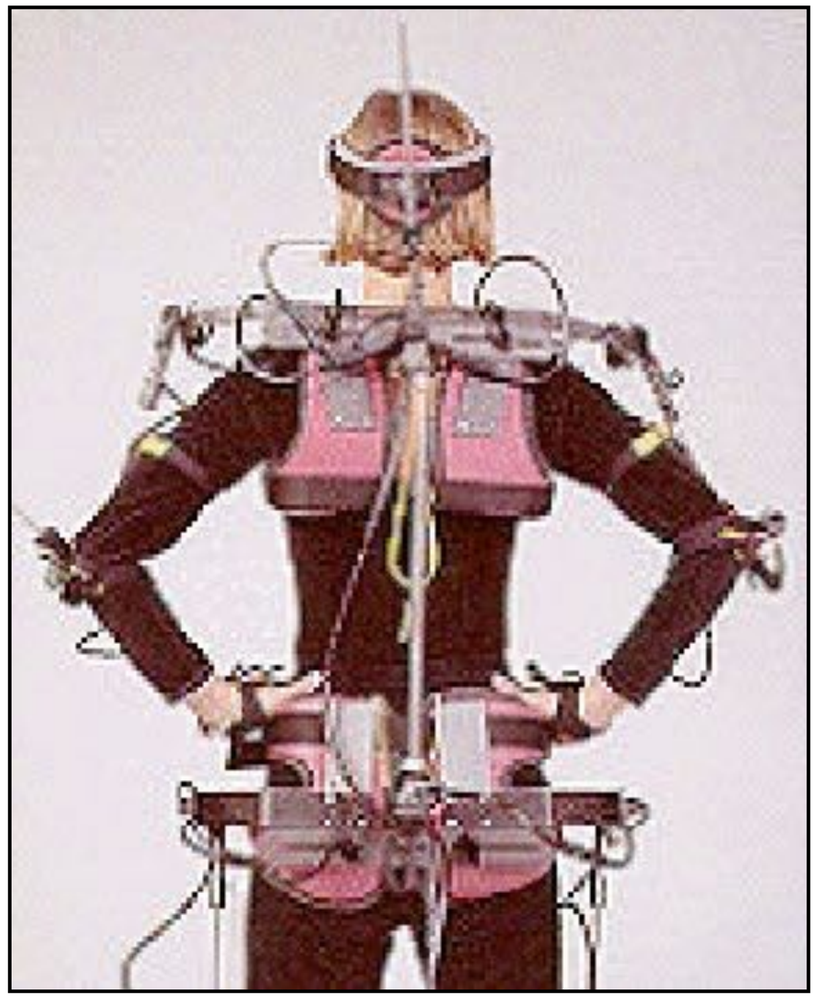
Data-driven approach to creating animation sequences

- Record real-world performances
- Extract pose as a function of time from raw data

Motion Capture Equipment

Optical
(More on following slides)

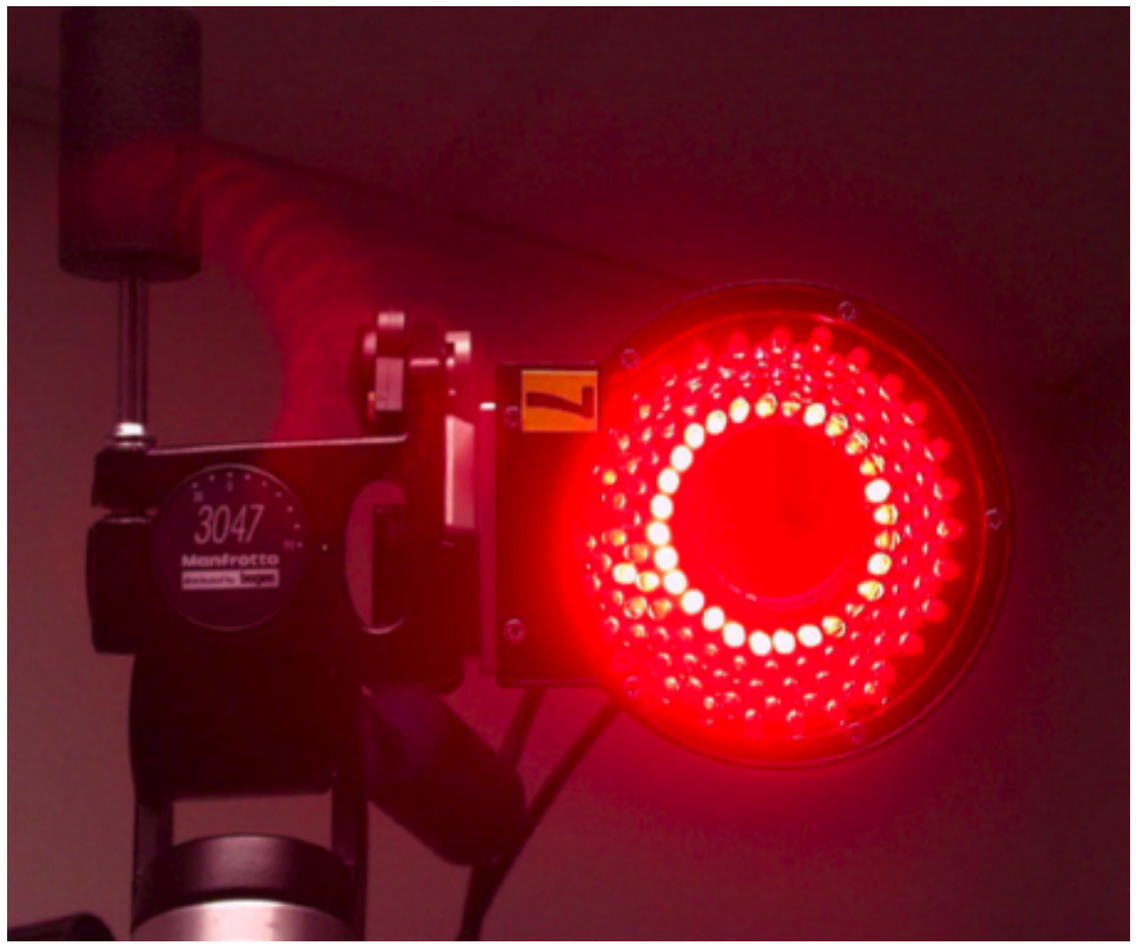
Magnetic
Sense magnetic fields to infer position / orientation.
Tethered.



Mechanical
Measure joint angles directly.
Restricts motion.

Optical Motion Capture

Retroflective markers attached to subject

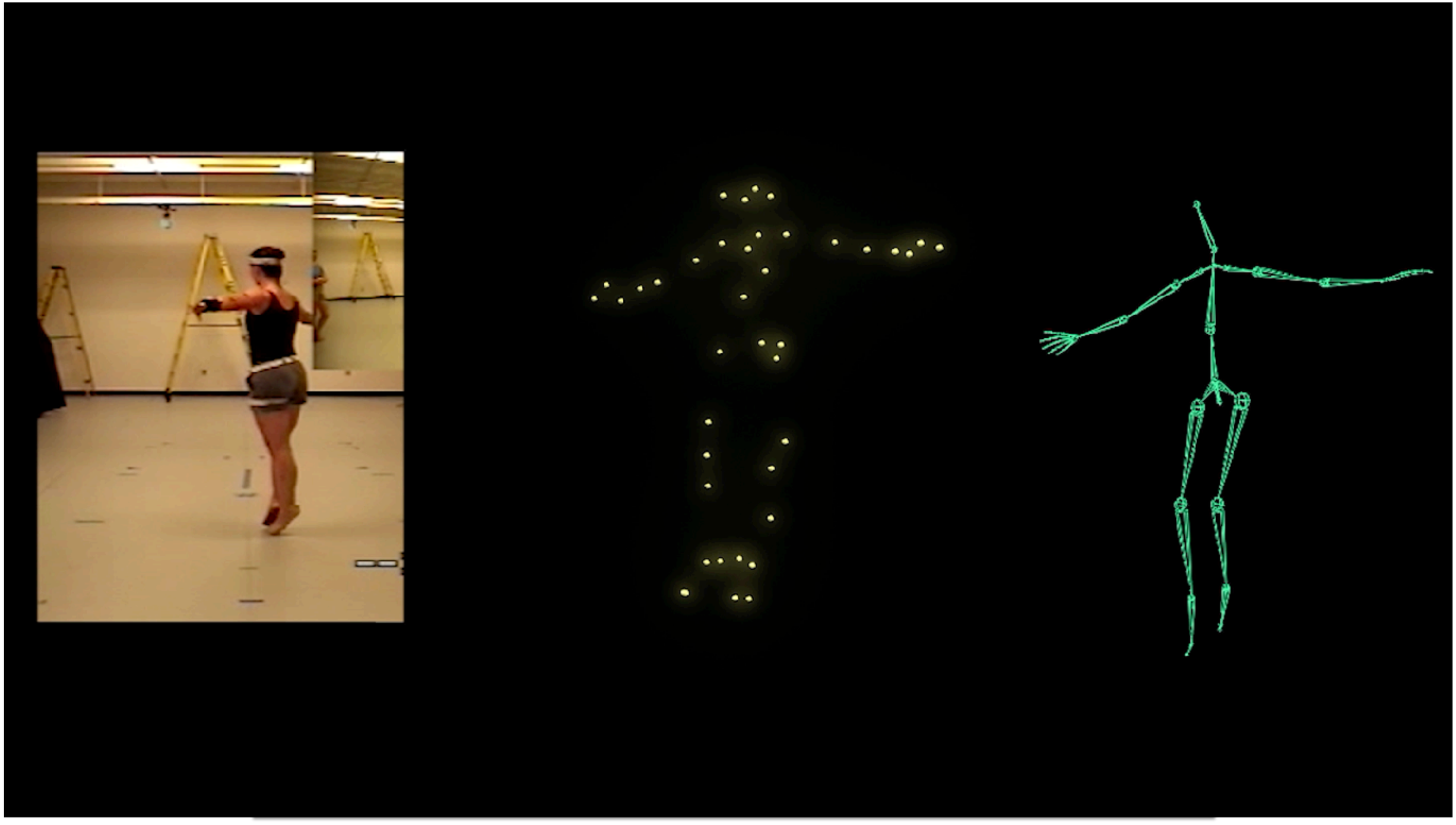


IR illumination and cameras

- Markers on subject
- Positions by triangulation from multiple cameras
- 8+ cameras, 240 Hz, occlusions are difficult

Slide credit: Steve Marschner

Motion Capture



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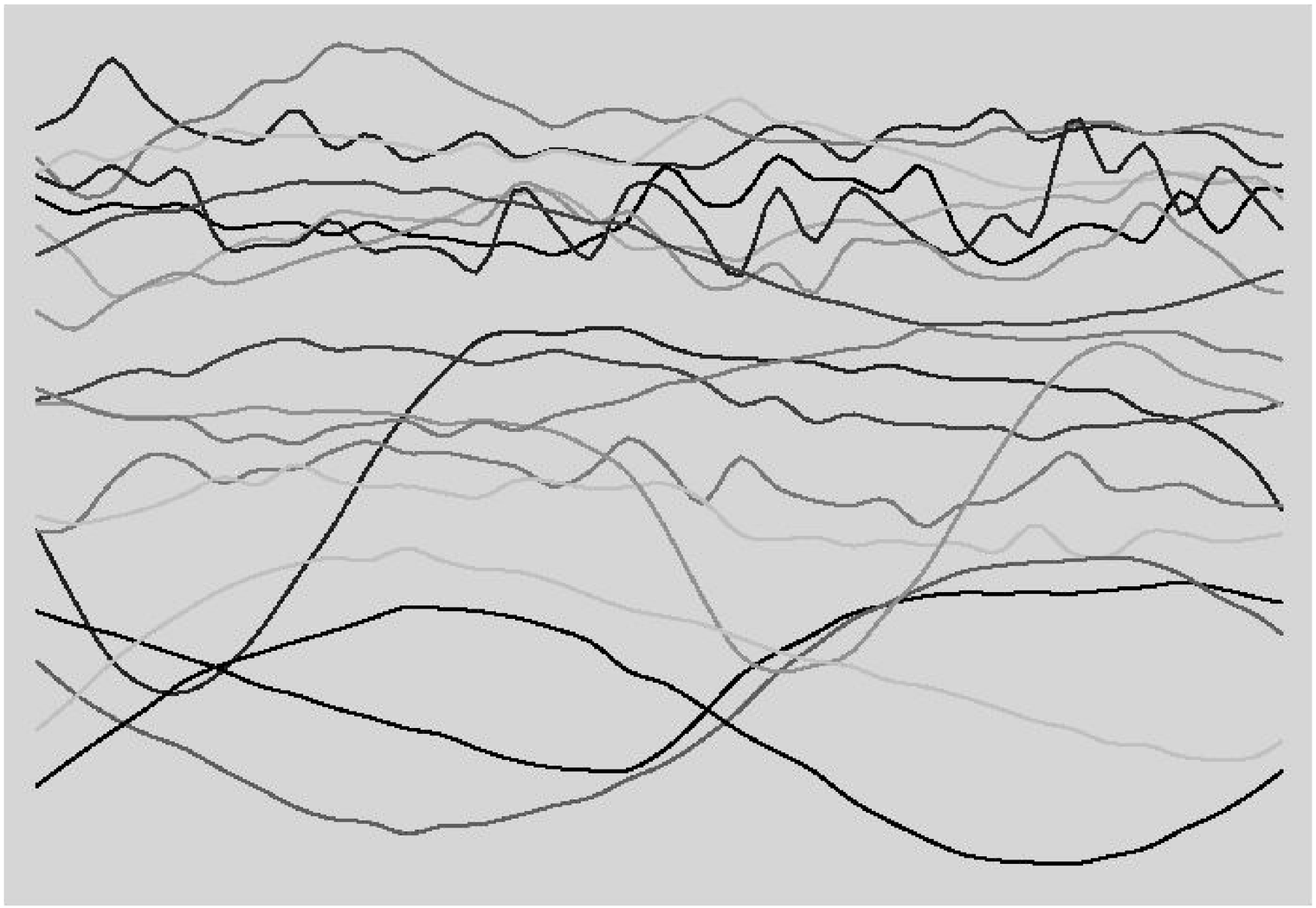
Motion Data



Subset of motion curves from captured walking motion.

From Witkin and Popovic, 1995

Motion Capture Pros and Cons

Strengths

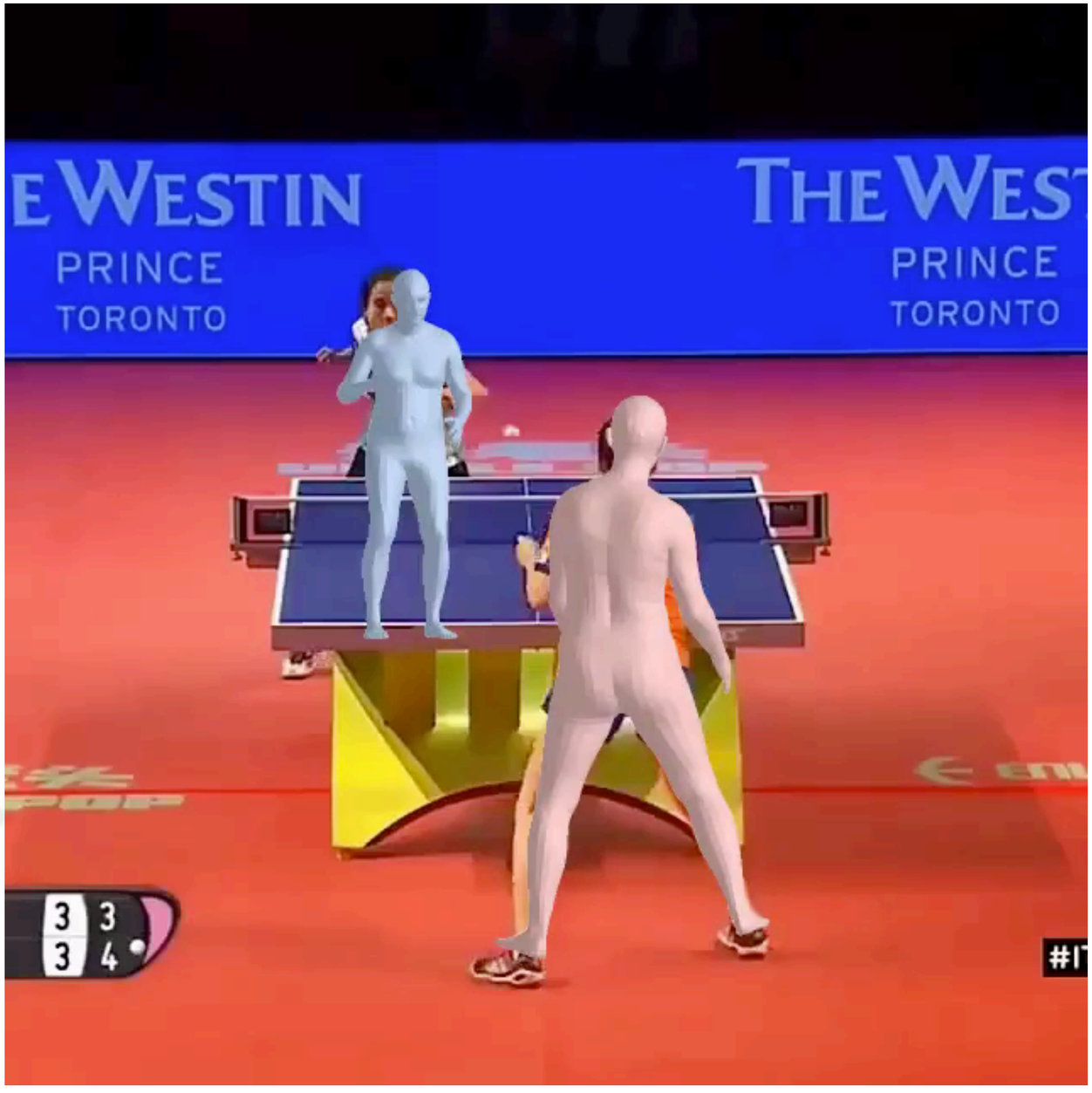
- Can capture large amounts of real data quickly
- Realism can be high

Weaknesses

- Complex and costly set-ups
- Captured animation may not meed artistic needs, requiring alterations

Markerless Motion Capture

Kanazawa et al. 2018



Kanazawa et al. 2019

Ren Ng
Presented by David McAllister

Acknowledgments

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